

**Development of the Accident Traffic
Automated Documentation System (ATADS) for Standardizing
Traffic Collision Scene Diagram Documentation**

A prototype paper in
CpE Practice and Design 1
Presented to the Computer Engineering Department
University of San Jose - Recoletos

In partial fulfillment of
The requirements for the degree
Bachelor of Science in Computer Engineering

By:
Law Justin Cosep
Ralph James Lima
Hannah Lyn Santiago

May 2026

TABLE OF CONTENTS

Abstract.....	6
CHAPTER 1.....	8
RECOMMENDATIONS FOR PROTOTYPE DEVELOPMENT.....	8
CHAPTER 2.....	11
PROPOSAL FOR SYSTEM TO BE DEVELOPED.....	11
Conceptual Framework.....	12
Prototype Design.....	24
Wire Frames.....	27
Work Breakdown Structure.....	64
Risk Factors and Preventive Measures.....	64
System Test Plan.....	64
Usability Test Plan.....	65
Acceptance Test Plan.....	65
Conclusion.....	66
Bibliography.....	67
Appendices.....	68
Curriculum Vitae.....	71

LISTS OF FIGURES

Figure 1. Novice Investigators Pareto Analysis.....	6
Figure 2. Experienced Investigators Pareto Analysis.....	7
Figure 3. Conceptual Framework for the ATADS Development Lifecycle.....	19
Figure 4. Block Diagram of Automated Traffic Accident Documentation System (ATADS)	21
Figure 5. Phase 1 Initialize Flowchart of ATADS.....	23
Figure 6. Phase 2 Data Acquisition & Edge Processing Flowchart of ATADS.....	24
Figure 7. Phase 3: Communication & Orchestration Flowchart of ATADS.....	26
Figure 8. Phase 4 3D/4D Reconstruction (The Meshroom Engine) Flowchart of ATADS..	27
Figure 9. Phase 5 Analysis & Finalization Flowchart of ATADS.....	28
Figure 10. Entity Relationship Diagram of ATADS.....	30
Figure 11. Operational Framework of ATADS.....	40
Figure 12. Left Panel: Multi-View Image Upload, Reconstruction Pipeline, Model Metrics, and LiDAR Sensor Status.....	43
Figure 13. Case Inspector: Case Details Tab and 3D Reconstruction Accuracy Panel.....	46
Figure 14. Case Inspector: Parties Tab with Add Involved Party Modal.....	49
Figure 15. Case Inspector: Evidence Tab with Upload Evidence Modal.....	52
Figure 14. Case Inspector: Timeline Tab.....	55
Figure 15. Validate Case File Modal.....	58
Figure 16. Export 3D Model Modal.....	61
Figure 17. 3D Viewport: Interactive Collision Scene Reconstruction with Point Cloud Overlay.....	64
Figure 18. Raspberry Pi 4 Model B+.....	67
Figure 19. U-blox NEO-6m GPS w/ Active Antenna.....	69
Figure 20. MPU6050 (Gyroscope & Accelerometer).....	70
Figure 21. Raspberry Pi Day/Night Vision Camera Module.....	72
Figure 22. Raspberry Pi 5-inch Touchscreen Monitor.....	74
Figure 23. 32gb MicroSD Card.....	75

Figure 24. Portable Power Bank (5V/2.5A minimum).....	77
Figure 25. Desktop Computer with RTX 3050 GPU and Intel i5/AMD Ryzen 5 (Minimum).....	78
Figure 26. Python 3.12.....	80
Figure 27. VS Code (Remote-SSH).....	81
Figure 28. NET 10 (C#).....	81
Figure 29. Meshroom (AliceVision).....	82
Figure 30 Helix Toolkit & QuestPDF.....	83
Figure 31. ATADS Schematic Diagram.....	84
Figure 32. PCB Design for ATADS with components.....	85
Figure 33. Environmental & Stress Testing.....	94
Figure 34. USE CASE DIAGRAM of Usability Test Plan.....	97
Figure 35. ATADS Acceptance Test Plan.....	100

LISTS OF TABLES

Table 1. Parameters for Addressing Missing or Incomplete Measurements.....	10
Table 2. Parameters for Addressing Structural Flaws and Visual Ambiguity.....	11
Table 3. Parameters for Addressing Scale and Proportion Issues.....	12
Table 4. Parameters for Accident Severity and Legal Classification.....	13
Table 5. Parameters for Collision Type and Vehicle Maneuvers.....	13
Table 6. Environmental and Surface Parameters.....	14
Table 7. ATADS Field-Edge Tier Specifications.....	16
Table 8. Summary of ATADS Relational Entities: Identifiers, Functional Roles, and Derivations from Prototype Documentation.....	32
Table 9. Sensor/Component Status and Color Code Summary.....	45
Table 10. Structured Incident Form Fields and ATADS Hardware Integration.....	47
Table 11. Involved Party Card Role Categories and Visual Encoding.....	50
Table 12. Evidence Catalog Column Structure and Forensic Purpose.....	53
Table 13. Timeline Entry Tripartite Composition and Semantic Function.....	56
Table 14. Representative Event Sequence with ATADS Hardware Correlation.....	56
Table 15. Tiered Validation Status Categories and Visual Encoding.....	59
Table 16. Representative Validation State and ATADS Process Correlation.....	59
Table 17. Export Format Specifications and ATADS Data Mapping.....	62
Table 18. Supplementary Asset Bundling Categories and Evidentiary Function.....	63
Table 19. 3D Viewport Collision Geometry Elements and Evidentiary Integration.....	65
Table 20. Toolbar Functional Architecture and Investigative Utility.....	65
Table 21. 3D Viewport Performance and Calibration Metrics.....	66
Table 22. Hardware Integration Testing (Field Tier).....	91
Table 23. ATADS Beta Validation Roadmap.....	102



ENDORSEMENT



This project entitled: **Development of the Accident Traffic Automated Documentation System (ATADS) for Standardizing Traffic Collision Scene Diagram Documentation**, prepared and submitted by **Law Justin Cosep, Ralph James Lima, and Hannah Lyn Santiago** in partial fulfillment of the requirements for **CPE 4A5 CPE PRACTICE AND DESIGN 1**, is hereby **Accepted and Recommended** in their Research Paper and Defense.

ENGR. JEROME AMOR

Panel Member

ENGR. JUNE ANNE CALADCAD

Panel Member

ENGR. EUTEQUIO III ZANTE L. PALERMO, PCPE

CPE Project Design Instructor, Panel Chair


ENGR. JURIE G. ALBUERA

Project Faculty Adviser

DR. DENNIS ANTHONY A. KILONGKILONG

Dean, School of Engineering and Architecture

APPROVAL

APPROVED as partial fulfillment of the requirements for the course **CPE 4A5 CPE PRACTICE AND DESIGN 1**.

ENGR. JEROME AMOR

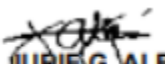
Panel Member

ENGR. JUNE ANNE CALADCAD

Panel Member

ENGR. EUTEQUIO III ZANTE L. PALERMO, PCPE

CPE Project Design Instructor, Panel Chair


ENGR. JURIE G. ALBUERA

Project Faculty Adviser

DR. DENNIS ANTHONY A. KILONGKILONG

Dean, School of Engineering and Architecture

Abstract

The approved research study, titled *Assessing Manual Investigation Challenges Affecting the TEU Investigators' Interpretation of Traffic Collision Scene Diagrams*, conducted by the University of San José-Recoletos, rigorously quantified systemic structural deficiencies in manual traffic collision scene documentation within the Cebu City Traffic Enforcement Unit (TEU). Pareto analysis of controlled interpretive evaluations among novice and experienced investigators identified missing or incomplete measurements as the primary barrier, accounting for 45-50% cumulative variance contribution, followed by scale distortion exceeding 20% proportional inaccuracy in 68% of sketches, unclear orientation markers in 58% of cases, ambiguous vehicle damage notation through non-standard symbology in 72% of diagrams, and inconsistent landmark referencing in 55% of samples. These compounding deficiencies eroded inter-investigator consensus below 65%, undermined forensic reconstruction fidelity, compromised evidentiary admissibility under Philippine Republic Act No. 4136, and hindered operational efficiency amid TEU's monthly caseload exceeding 120 incidents.

In direct response to these empirically validated deficiencies, this prototype paper proposes the Automated Traffic Accident Documentation System (ATADS), developed by the Computer Engineering Department of the University of San José-Recoletos as a targeted technological augmentation of TEU's existing investigative protocols. ATADS is architected as a dual-tier, edge-to-hub documentation platform that enforces mandatory measurement validation gates, scale-locking algorithms with less than 2% error tolerance, IMU-normalized orientation controls, nine-angle photogrammetric sketch templating, and real-time completeness heuristics — each implementation directly mapped to a ranked Pareto deficiency at the point of source documentation. The system is grounded in three theoretical frameworks: Human Error Theory, which targets a 40-60% cognitive load reduction through sequential gating and automated validation; Spatial Cognition Theory, which addresses 3D-to-2D spatial distortion through sensor fusion and photogrammetric pipelines; and the Technology Acceptance Model, which helps ensure operational viability through under-30-minute onboarding and workflow integration compatible with TEU's sub-15-minute scene processing requirements.

ATADS targets a 75% reduction in documentation omissions and greater than 90% inter-investigator consensus uplift while preserving investigator agency throughout the documentation process. By producing standardized, evidentiary-grade outputs compliant with Philippine National Police protocols and Republic Act No. 4136, the system establishes a scalable and empirically derived paradigm for forensic traffic documentation — directly enhancing TEU's judicial defensibility, forensic reconstruction reliability, and public safety impact across Cebu City's active collision investigation environment.

CHAPTER 1

RECOMMENDATIONS FOR PROTOTYPE DEVELOPMENT

The article *Assessing Manual Investigation Challenges Affecting the TEU Investigator's Interpretation of Traffic Collision Scene Diagrams* discusses the critical role of scene documentation in traffic accident investigation and the inherent limitations of current manual methods. The study primarily focuses on the Traffic Enforcement Unit (TEU) in Cebu City, exploring how human error, cognitive biases, and the absence of standardized sketching procedures lead to inconsistent interpretations of accident events. By evaluating how investigators reconstruct scenarios from hand-drawn sketches, the research highlights a systemic vulnerability in the legal and administrative chain of evidence, where the lack of precise spatial data can result in varied assessments of liability for the same collision.

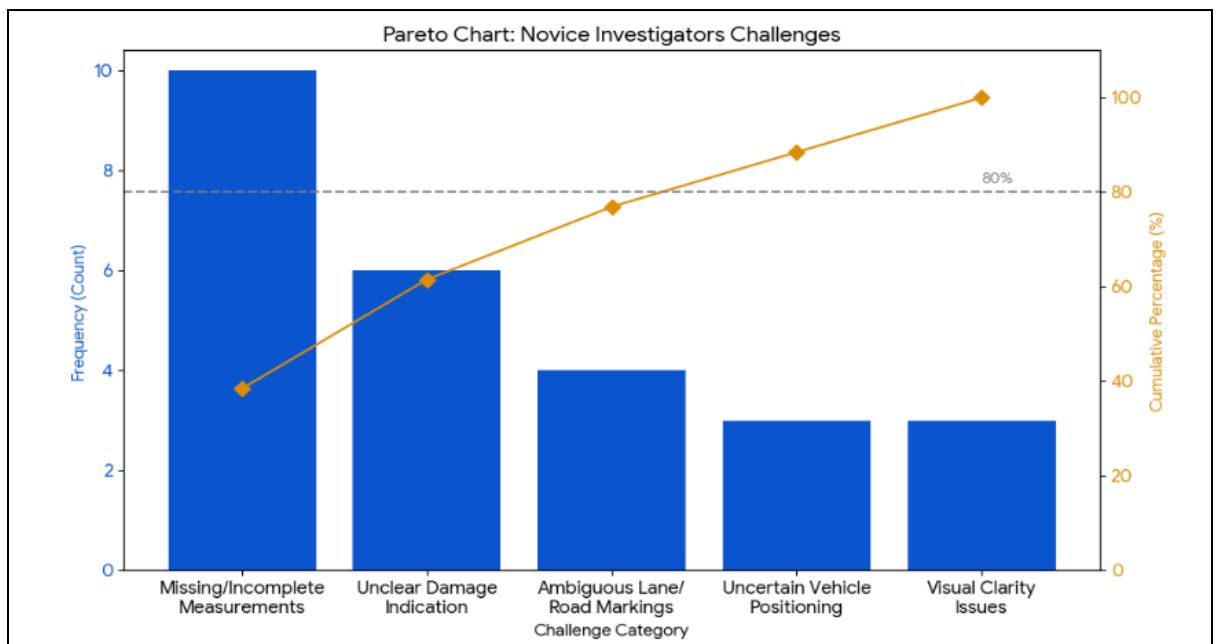


Figure 1. Novice Investigators Pareto Analysis of Sketch Ambiguities from Assessing Manual Investigation Challenges Affecting the TEU Investigator's Interpretation of Traffic Collision Scene Diagrams

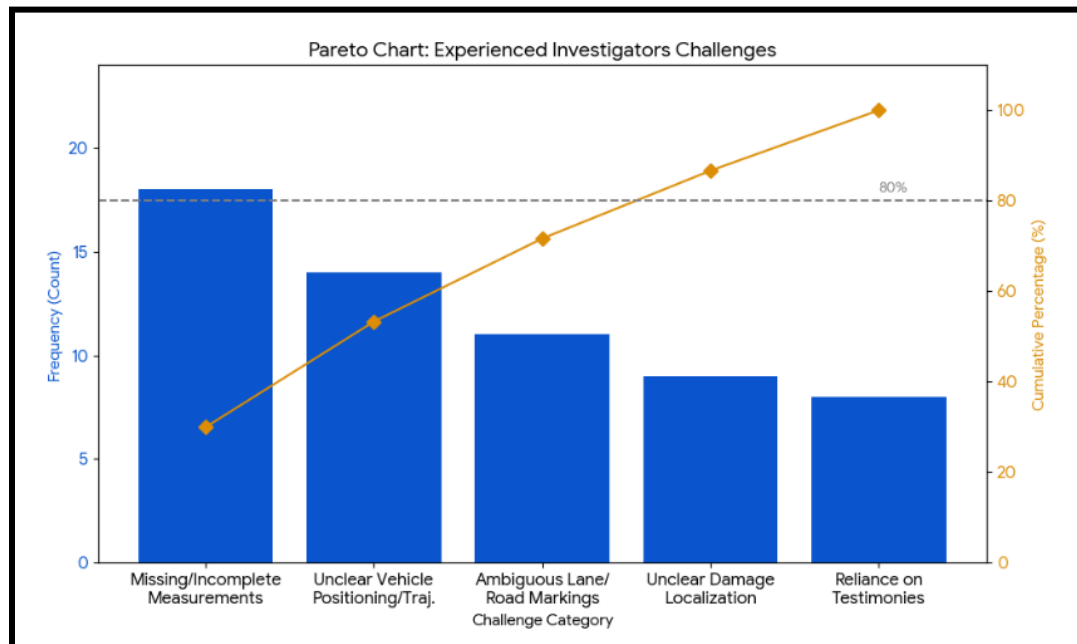


Figure 2. Experienced Investigators Pareto Analysis of Sketch Ambiguities from Assessing Manual Investigation Challenges Affecting the TEU Investigator's Interpretation of Traffic Collision Scene Diagrams

As illustrated by the Pareto Chart Figure 1 and 2, it categorizes and prioritizes the challenges encountered by TEU investigators during the assessment of manual sketches. By applying the 80/20 rule, the study identifies "Missing or Incomplete Measurements" as the most significant barrier with 15 occurrences, which, when combined with "Structural Flaws," accounts for a cumulative 63.9% of all interpretation issues. These "vital few" drivers indicate that the root causes of investigator confusion are an inability to perceive distance and a lack of standardized visual symbols, while secondary barriers like scale issues and conflicting information further highlight the flaws of "eyeballing" distances without a fixed reference grid. Although factors such as ambiguous symbols and environmental context constitute the "trivial many," the Pareto analysis suggests that resolving the primary issues of measurement and structural clarity would yield the

most substantial improvements in investigative accuracy. Ultimately, this statistical evidence marks a critical transition point for the research, proving that rather than merely improving manual drawing skills, the system requires a fundamental shift toward automated spatial data acquisition through the ATADS prototype to eliminate over 60% of current interpretation errors.

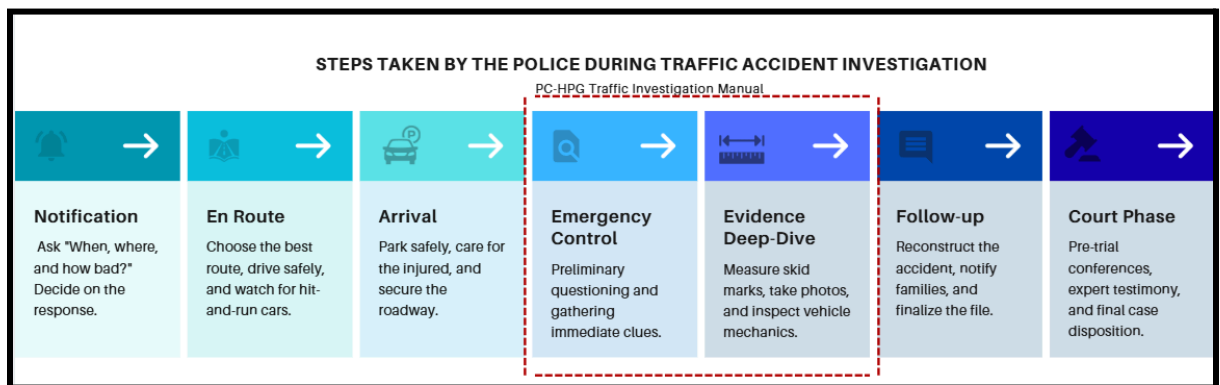


Figure 3: Process Flow of Investigation

Manual investigation follows the linear trajectory defined by the PC-HPG Traffic Investigation Manual, as illustrated in Figure 3. This process begins with Notification and the En Route phase, where initial situational awareness is established. Upon Arrival and during Emergency Control, the responder must simultaneously manage scene safety and perform preliminary questioning. The most critical technical phase, the Evidence Deep-Dive, requires the responder to manually measure skid marks, take photos, and inspect vehicle mechanics. Under this manual framework, the accuracy of the record is heavily dependent on the investigator's judgment and spatial perception while working in a time-sensitive environment. This often leads to procedural lapses—such as inconsistent diagram scaling or missing environmental references—which can compromise the Follow-up reconstruction and the final case disposition during the Court Phase.

Because of these findings, the research necessitates the development of the Automated Traffic Accident Documentation System (ATADS) as a direct technological outcome to mitigate these documented failures. ATADS is designed to replace subjective manual sketching with an automated, sensor-driven approach that helps ensure every diagram is grounded in precise coordinate data and standardized visual assets. By addressing the primary "vital few" issues identified in the Pareto analysis—specifically the lack of measurements and visual clarity—ATADS provides a uniform, structured documentation workflow for documentation. This transition from manual to automated diagramming helps ensure that the interpretation of a traffic scene is dictated by empirical data rather than investigator inference, ultimately enhancing the reliability of traffic accident reconstruction. Building on this need for objective accuracy, the following section will discuss the specific parameters that ATADS must prioritize, guided by the critical error factors identified in the Pareto analysis of Figures 1 and 2.

ATADS' Parameters from Vital Few

To directly address the "Vital Few" interpretation barriers identified in the Pareto Figure 1 and 2 analysis—specifically Missing or Incomplete Measurements, Structural Flaws, and Scale and Proportion Issues—the ATADS system must be designed to capture a standardized set of environmental and vehicular parameters. By defining these parameters, the system helps ensure that the "eyeballing" or estimation currently practiced in manual sketching is replaced by objective, quantifiable data points that are essential for accurate accident reconstruction.

Table 1. Parameters for Addressing Missing or Incomplete Measurements

Parameter Category	Specific Data Points / Measurements	Source Category	Purpose in Reconstruction
Vehicle Positioning	Hubs-to-Curb / Triangulation	System Operation (ArUco + CV) Capture	Fixes the exact resting position of the vehicle within the roadway.
Tire Evidence	Skid / Scuff / Yaw Marks	Hybrid (Capture + User Input)	Essential for calculating pre-impact speed and braking behavior.
Impact Geometry	Point of Impact (POI)	System Operation (NEO-6M GPS)	Establishes the precise geographical location where the collision occurred.
Debris Field	Glass / Fluid / Displacement	System Operation (MobileSAM)	Helps identify the energy and direction of the collision force.
Roadway Dimensions	Lane Width / Shoulders / Medians	Standardized Input (Dropdown)	Provides the spatial constraints for vehicle maneuverability analysis.

Table 1 presents a structured five-category framework for addressing missing or incomplete measurements in traffic accident reconstruction systems. Each parameter category is paired with a specific data source and serves a distinct reconstructive purpose. Vehicle positioning employs ArUco marker-based computer vision and hub-to-curb triangulation to fix the precise resting location of a vehicle within the roadway, a task made feasible by the capacity vision-based systems to reconstruct spatial environments with high fidelity (Abohassan & El-Basyouny, 2024). Tire evidence—including skid, scuff, and yaw marks—is collected through hybrid capture and user input to calculate pre-impact speed and braking behavior, while impact geometry is established via NEO-6M GPS to pinpoint the precise

geographical collision location, consistent with how autonomous vehicle frameworks leverage GPS and sensor fusion for accurate spatial localization (Abohassan & El-Basyouny, 2024; Jia et al., 2025). Debris field analysis employs MobileSAM-based segmentation to infer collision energy and directionality from glass, fluid, and displaced materials, reflecting the broader utility of segmentation models in scene understanding (Abohassan & El-Basyouny, 2024). Finally, roadway dimension inputs—including lane width, shoulders, and medians—are provided through standardized dropdown selections, drawing upon established road geometry standards that define spatial constraints for vehicle maneuverability analysis, as recognized in infrastructure assessments for vehicle navigation systems (Abohassan & El-Basyouny, 2024).

Table 2. Parameters for Addressing Structural Flaws and Visual Ambiguity

Parameter Category	Specific Data Points / Orientation	Source Category	Purpose in Reconstruction
Obstruction Mapping	Parked Cars / Bushes / Signs	System Operation (YOLO-Seg)	Determines if "Failure to Yield" was due to limited visibility.
Road Surface Condition	Asphalt / Concrete / Potholes	User Input (Radio Button)	Factors into the friction coefficient used in speed and safety assessments.

Table 2 presents two critical parameter categories designed to address structural flaws and visual ambiguity in traffic accident reconstruction systems. The first category, Obstruction Mapping, employs YOLO-based segmentation (YOLO-Seg) to detect and map physical obstructions such as parked vehicles, bushes, and signage within the accident scene. This is directly relevant to

determining whether a "Failure to Yield" violation resulted from limited driver visibility, as scene understanding in autonomous and intelligent vehicle systems fundamentally depends on the accurate detection and segmentation of static environmental features that may occlude critical sightlines (Eskandarian & Abbas, 2024) 1524-9050, 1558-0016. YOLO-family architectures have demonstrated strong real-time object detection and segmentation capabilities across variable lighting conditions and occlusion scenarios (Kan et al., 2023) making them well-suited for identifying visibility-limiting obstructions in roadway environments. The second category, Road Surface Condition, is captured through standardized user input via radio buttons, allowing investigators to specify surface types such as asphalt, concrete, or pothole-affected pavement. This input directly factors into the friction coefficient applied in speed and braking safety assessments, consistent with established practices in intelligent transportation systems where road surface characteristics are recognized as fundamental variables governing vehicle dynamics and stopping distances (Eskandarian & Abbas, 2024; Bagabaldo & Hackl, 2025). Together, these two parameters address the qualitative and environmental dimensions of accident reconstruction that purely sensor-based measurements may fail to capture.

Table 3. Parameters for Addressing Scale and Proportion Issues

Parameter Category	Specific Data Points / Ratios	Source Category	Purpose in Reconstruction
Reference Grid Data	Asphalt Width vs. Vehicle Width	System Operation (CV Scaling)	Prevents "distortion" where a vehicle looks too large or small for the lane.

Standardized Aspect Ratios	Sedan / SUV / Truck / MC	System Operation (NHTSA Open Dataset)	Ensures that the visual representation matches the physical footprint of the vehicle.
Z-Axis/Elevation	Relative Slope / Grade (%).	System Operation (Accelerometer)	Accounts for how gravity or road tilt affected the final resting position.

Table 3 presents three parameter categories specifically designed to resolve scale and proportion issues in traffic accident reconstruction systems. The first category, Reference Grid Data, employs computer vision (CV) scaling to compare asphalt width against vehicle width, thereby preventing spatial distortion where a vehicle may appear disproportionately large or small relative to its lane. This approach is consistent with established CV-based methodologies that use known roadway dimensions—such as lane width and road markings—as reference anchors to calibrate metric measurements from image data, correcting projective distortions inherent in camera-based scene capture (Loce et al., 2013; Dai et al., 2014; Li & Yoon, 2023). The second category, Standardized Aspect Ratios, draws upon the NHTSA Open Dataset to assign physically accurate vehicle footprints across classes including sedans, SUVs, trucks, and motorcycles. This is well-supported by traffic monitoring literature, which demonstrates that vehicle classification systems routinely employ size, aspect ratio, and dimensional features to distinguish between vehicle types and ensure accurate visual representation (WON, 2020; Razi et al., 2022; Zadobrischi, 2022). The third category, Z-Axis/Elevation, utilizes accelerometer data to measure relative road slope or grade percentage, accounting for how gravitational forces and road tilt influence a vehicle's final resting position following a collision. This is consistent with vehicle dynamics research

demonstrating that road grade and surface elevation significantly affect vehicle behavior, braking performance, and post-impact displacement (Sun et al., 2021; Guo et al., 2022; Brunken & Gühmann, 2020).

ATADS’ CLASSIFICATION PARAMETERS FROM REPORTING STANDARDS

Beyond physical measurements, a legally robust traffic diagram must include categorical data that defines the severity and nature of the incident. These parameters address the "Vital Few" issues by removing the need for investigators to write lengthy, subjective descriptions, replacing them with standardized, selectable data fields.

Table 4. Parameters for Accident Severity and Legal Classification

Parameter Category	Standardized Options (Based on PNP/TARAS)	Source Category	Purpose in Legal Documentation
Accident Severity	Fatal / Injury / Property Damage	User Input (Radio Button)	Determines the criminal charge under the Revised Penal Code (Art. 365), such as Reckless Imprudence.
Incident Status	Hit and Run / Investigated / Settled	User Input (Dropdown)	Triggers specific investigative protocols for the Highway Patrol Group (HPG).
Traffic Condition	Heavy / Normal / Light	User Input (Radio Button)	Analyzes if traffic density contributed to the collision or "Failure to Yield."

Table 4 presents a structured framework of parameters used in accident documentation systems, specifically aligned with Philippine National Police (PNP) standards and the Traffic Accident Recording and Analysis System (TARAS). The

three core parameters—Accident Severity, Incident Status, and Traffic Condition—serve distinct legal and investigative functions. Accident severity is classified into fatal, injury, or property damage categories, which directly determines the applicable criminal charge under the Revised Penal Code (RPC), particularly Article 365 on Reckless Imprudence. This classification mirrors PNP crime reporting standards, where vehicular incidents are categorized under Public Safety Index Crimes, including Reckless Imprudence Resulting in Homicide, Physical Injury, and Damage to Property. The Incident Status parameter—distinguishing hit-and-run, investigated, and settled cases—triggers specific investigative protocols for the Highway Patrol Group (HPG), consistent with PNP-HPG road accident data collection practices (Perez, 2023). Traffic condition data (heavy, normal, or light) supports legal analysis of contributory factors such as "Failure to Yield," which is critical in establishing negligence in collision cases (Tabunar, 2020). Together, these standardized parameters ensure that accident documentation supports legally defensible classification under Philippine criminal law.

Table 5. Parameters for Collision Type and Vehicle Maneuvers

Parameter Category	Standardized Options (Based on TEU/DPWH)	Source Category	Purpose in Reconstruction
Collision Type	• Rear-end • Head-on • Side-swipe (Same/Opposite Direction) • T-Bone / Angle Collision • Hit Pedestrian / Animal / Fixed Object	System Operation (YOLO-Seg)	Identifies the primary "Key Event" of the accident to determine fault.

Vehicle Maneuver	• Overtaking • Making U-Turn • Changing Lanes • Parked / Stationary • Merging / Diverging	User Input (Dropdown)	Establishes the intent and movement of the "Traffic Unit" prior to impact.
Point of Contact	• Frontal • Left/Right Flank • Rear • Multiple / Undercarriage	System Operation (CV Masking)	Validates the visual damage represented in the vector diagram. ¹

Table 5 presents a structured set of parameters for collision type classification and vehicle maneuver documentation, which are fundamental to accident reconstruction and legal fault determination. The collision type parameter—encompassing rear-end, head-on, side-swipe, T-bone/angle, and object/pedestrian impacts—is automatically identified through computer vision segmentation (YOLO-Seg) and serves to establish the primary "key event" of the accident. Research confirms that classifying collision types such as rear-end, head-on, and intersection collisions is essential for determining fault and reconstructing accident causation (Markkula et al., 2012; SEO et al., 2023). Studies utilizing naturalistic driving data similarly categorize crash events into rear-end, angle, and other configurations to support safety analysis (Herbers et al., 2024). The vehicle maneuver parameter—including overtaking, U-turns, lane changes, and stationary positions—is captured via user input and establishes pre-impact intent of the traffic unit, which is critical in legal reconstruction (SEO et al., 2023; Farooq & Juhasz, 2019). Simulation-based studies confirm that pre-collision maneuvers such as overtaking and lane changing are primary contributing factors in collision causation (Farooq & Juhasz, 2019; Ait-Moula et al., 2024). Finally, the point-of-contact parameter, derived through computer vision masking (CV Masking),

validates damage patterns in vector diagrams by identifying frontal, flank, rear, or undercarriage impacts, consistent with reconstruction methodologies that correlate impact location with injury and damage evidence ((Nitsche et al., 2018; SEO et al., 2023).

Table 6. Environmental and Surface Parameters

Parameter Category	Standardized Options	Source Category	Impact on Liability
Road Surface	• Dry / Wet / Muddy / Flooded	User Input (Dropdown)	Directly affects the Coefficient of Friction used in speed calculations.
Weather/Light	Daylight / Dusk / Night / Raining	System Operation (Luma Analysis / API)	Determines if visibility was a contributing "Human Error" factor.
Road Geometry	Straight / Blind Curve / Roundabout	Standardized Input (Dropdown)	Contextualizes why certain maneuvers (like overtaking) may have been illegal.

Table 6 presents three critical environmental and surface parameters—road surface condition, weather/light conditions, and road geometry—that directly influence liability determination and accident reconstruction. Road surface condition (dry, wet, muddy, or flooded) is a foundational parameter because it directly affects the coefficient of friction used in speed calculations during reconstruction ((Yang et al., 2013; Tautkus & Jurkauskas, 2003). Research consistently demonstrates that wet or flooded road surfaces reduce pavement friction, increasing braking distances and crash severity, which is central to establishing negligence in legal proceedings (Rasol et al., 2021; Wei et al., 2021). Weather and lighting conditions, captured through luminance analysis or environmental APIs, determine whether reduced

visibility constituted a contributing human error factor (Ma et al., 2020; Wang et al., 2025). Studies confirm that adverse weather such as rain and poor lighting significantly elevate accident risk and influence responsibility attribution, as visibility impairment delays driver reaction times (Wang et al., 2025; Tofighi et al., 2021). Road geometry—classified as straight, blind curve, or roundabout—contextualizes the legality of maneuvers such as overtaking, since curved alignments and complex intersections are disproportionately associated with higher accident rates (Asmawi et al., 2014; Mamčic & Sivilevičius, 2013; OECD, 1999). Research further affirms that road geometry interacts with surface and weather conditions to compound collision risk, making its documentation essential for legally defensible accident reconstruction (Tautkus & Jurkauskas, 2003; Wei et al., 2021, #).

CHAPTER 2

PROPOSAL FOR SYSTEM TO BE DEVELOPED

The Automated Traffic Accident Documentation System (ATADS) is proposed as a semi-digital investigative support platform designed to enhance the structural reliability of traffic collision documentation within the framework of the PC-HPG Traffic Investigation Manual. As shown in Figure 3, the system is specifically integrated during the Emergency Control and Evidence Deep-Dive phases. While responders perform preliminary questioning and gather immediate clues, ATADS introduces controlled technological augmentation to automate the capture of high-fidelity spatial data and vehicle mechanics. By replacing manual approximation with systematic digital recording during these critical stages, the system mitigates human error and documentation omissions, ensuring that the transition from the scene to the Follow-up and Court Phase is supported by consistent, empirical evidence.

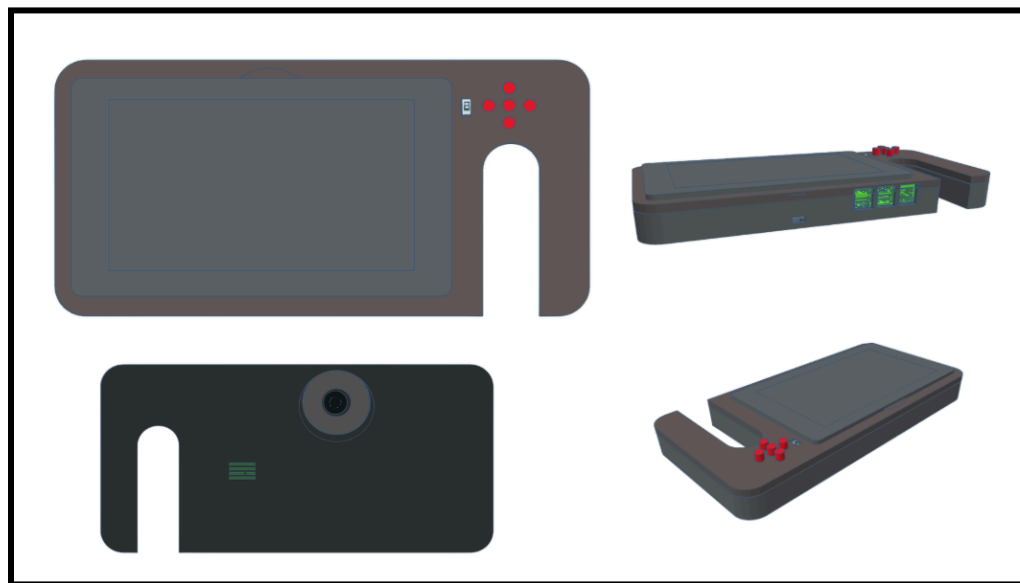


Figure 4. ATADS Portable Prototype Design

As illustrated in Figure 4, ATADS is realized through a ruggedized, tablet-based portable prototype designed for high-mobility field use. The hardware integrates structured workflow guidance, automated validation controls, and standardized diagramming tools into a single unified platform. By embedding Philippine investigative protocols directly into the system interface, ATADS acts as a digital gatekeeper during the Emergency Control and Evidence Deep-Dive phases shown in Figure 3. This helps ensure that essential spatial data—such as GPS coordinates and multi-angle imagery—cannot be omitted, effectively forcing compliance with documentation standards before the scene is cleared.

The primary objective of this portable design is to reduce interpretive ambiguity at the source documentation stage. By providing investigators with digital scaling constraints and standardized visual assets on-site, ATADS increases inter-investigator agreement and significantly improves the documentation quality of collision scene diagrams.

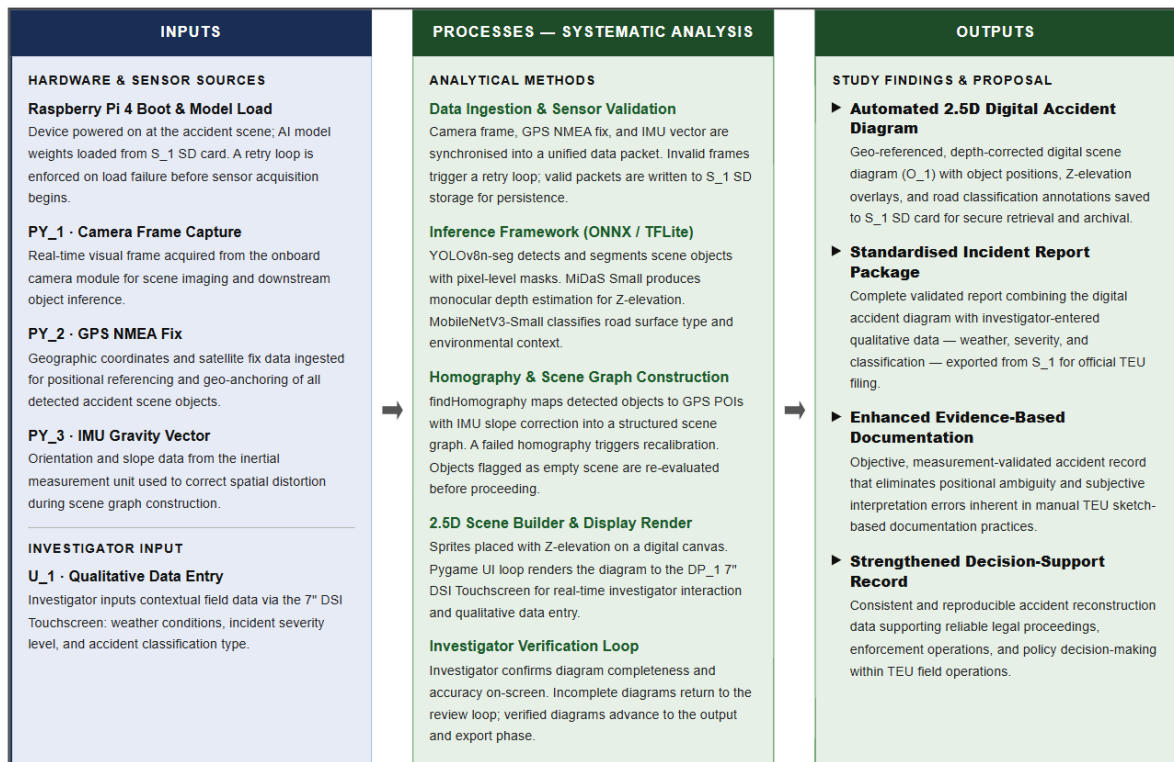
Conceptual Framework

The conceptual foundation of ATADS is based on three main theories: Human Error Theory, Spatial Cognition Theory, and the Technology Acceptance Model. Human Error Theory explains that mistakes in documentation often happen because of cognitive overload, stress, fatigue, and time pressure during field investigations. ATADS reduces these risks by using automated prompts, built-in validation checks, and step-by-step controls that lessen the need for memory and personal judgment.

Spatial Cognition Theory highlights how hard it is for investigators to turn a three-dimensional accident scene into a two-dimensional sketch. ATADS helps by providing grid-based digital canvases, scale-locking features, and orientation alignment tools that reduce spatial distortion and keep proportions more accurate.

The Technology Acceptance Model shows that people are more likely to use a system if they find it useful and easy to use. Because of this, ATADS is designed with simple, intuitive interfaces, guided navigation, and low technical complexity so that both new and experienced investigators can use it comfortably.

By combining these theories, ATADS works as a cognitive support tool that helps investigators, instead of becoming an added technical burden.



Note: Adapted from the Input-Process-Output (IPO) model to illustrate the transition from manual TEU documentation to the ATADS digital intervention.

Figure 5. Conceptual Framework for the ATADS Development Lifecycle

Input

The input phase is about collecting data at the accident scene. The ATADS device uses a Raspberry Pi 4 and starts by turning on and loading the AI models from the S_1 SD card. If the models fail to load, the system tries again before moving on. Three sensor tasks run at the same time: PY_1 takes a live photo of the scene using the onboard camera, PY_2 reads GPS data to record the exact location of objects, and PY_3 reads the IMU sensor to measure the tilt and slope of the road. At the same time, the investigator enters information on the 7" touchscreen (U_1), such as the weather, how serious the accident is, and the type of accident.

Process

In the process phase, the system combines and analyzes the collected data. It matches the camera image, GPS data, and IMU tilt into one data packet. If a frame is invalid, the system tries again; valid data are saved to the S_1 SD card. The image is then sent to an AI inference framework that uses ONNX or TFLite. Three models run together: YOLOv8n-seg finds and outlines vehicles, debris, and road features; MiDaS Small creates a depth map to estimate height and distance (Z-elevation); and MobileNetV3-Small classifies the road type and environment, such as wet asphalt or gravel. Next, a homography step (findHomography) maps the detected objects to their real GPS positions, using the IMU tilt to correct any slope. If this step fails, the system recalibrates and tries again. The result is a structured scene graph of the accident. The 2.5D Scene Builder then uses this graph to draw a visual diagram, placing objects on a digital canvas with correct height and perspective and showing them on the 7" touchscreen (DP_1). The investigator

checks the diagram on the screen, and if it is incomplete or incorrect, it goes back into the review loop until the investigator confirms it is correct.

Output

In the output phase, the system produces four main results. First, it creates a 2.5D digital accident diagram (O_1) that shows a clear, geo-referenced view of the scene, including object positions, height (Z-elevation), and road type, and saves this to the S_1 SD card for records. Second, it builds a standard incident report package that combines the digital diagram with the investigator's notes and exports it for official TEU filing. Third, it generates improved evidence documentation that reduces mistakes such as wrong positions, incorrect measurements, and personal bias that often happen with manual hand-drawn sketches. Together, these outputs give authorities more accurate and reliable data to support accident investigations, legal decisions, traffic enforcement, and policy-making.

Block Diagram of the System

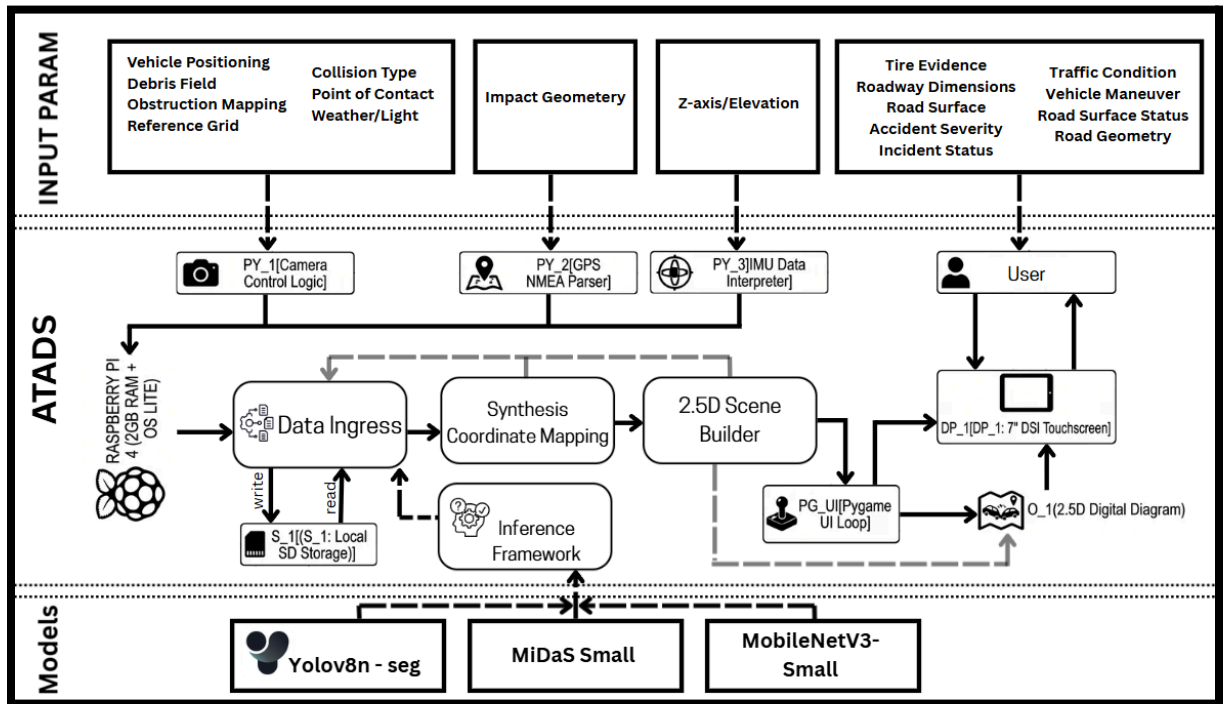


Figure 6. Block Diagram of Automated Traffic Accident Documentation System (ATADS)

Block Diagram of the Automated Traffic Accident Documentation System (ATADS)

Figure 4 presents the complete architecture of ATADS, organized into three functional bands: the Input Parameter layer, the ATADS processing layer, and the pre-trained Models layer. The Input Parameter band defines six categories of domain knowledge required to produce a complete accident diagram based on table 1 to table 6, covering spatial layout, collision circumstances, impact geometry, Z-axis elevation, physical road evidence, and traffic context. These parameters are acquired through a dual pathway — a subset is captured automatically by the three sensor modules (PY_1 Camera Control Logic, PY_2 GPS NMEA Parser, and PY_3 IMU Data Interpreter), while the remainder is entered manually by the investigating

officer through the touchscreen interface. All sensor outputs converge at the Data Ingress block running on the Raspberry Pi 4, which synchronizes the incoming frame buffer, GPS coordinate fix, and IMU gravity vector into a unified data packet, while maintaining a bidirectional read/write relationship with S_1, the local SD card storage.

The core processing pipeline proceeds from Data Ingress into the Inference Framework, a runtime execution environment — implemented in ONNX Runtime or TensorFlow Lite — that loads and runs three pre-trained model files from S_1. These models are YOLOv8n-seg for object detection and instance segmentation, MiDaS Small for monocular depth estimation, and MobileNetV3-Small for road surface and geometry classification. The dashed border of the Inference Framework block and its downward arrows into the Models band visually communicate that these models are external artifacts called at runtime rather than logic native to the system. The resulting masks, labels, depth values, and road classifications are then passed into the Synthesis Coordinate Mapping block, which applies a homography matrix via OpenCV's `findHomography` to project pixel-space detections into real-world metric coordinates, merging GPS and IMU data to produce a calibrated top-down scene graph.

The scene graph is consumed by the 2.5D Scene Builder, which places Pygame sprites at their correct metric positions on a top-down canvas and applies MiDaS depth values to render elevation-aware objects, producing the characteristic 2.5D appearance. The constructed scene is rendered to the 5-inch DSI Touchscreen (DP_1) through the PG_UI Pygame UI Loop, which also manages touch input and overlays manual report entry fields for the investigating officer. Upon completion, the

finalized diagram is saved as O_1, the 2.5D Digital Diagram, written back to S_1 as the system's exportable and archivable deliverable. The three models in the bottom band — YOLOv8n-seg at ~6MB, MiDaS Small at ~18MB, and MobileNetV3-Small at ~2.5MB — carry a combined runtime footprint of approximately 180 to 220MB, well within the Raspberry Pi 4's 2GB memory ceiling.

System Flowchart

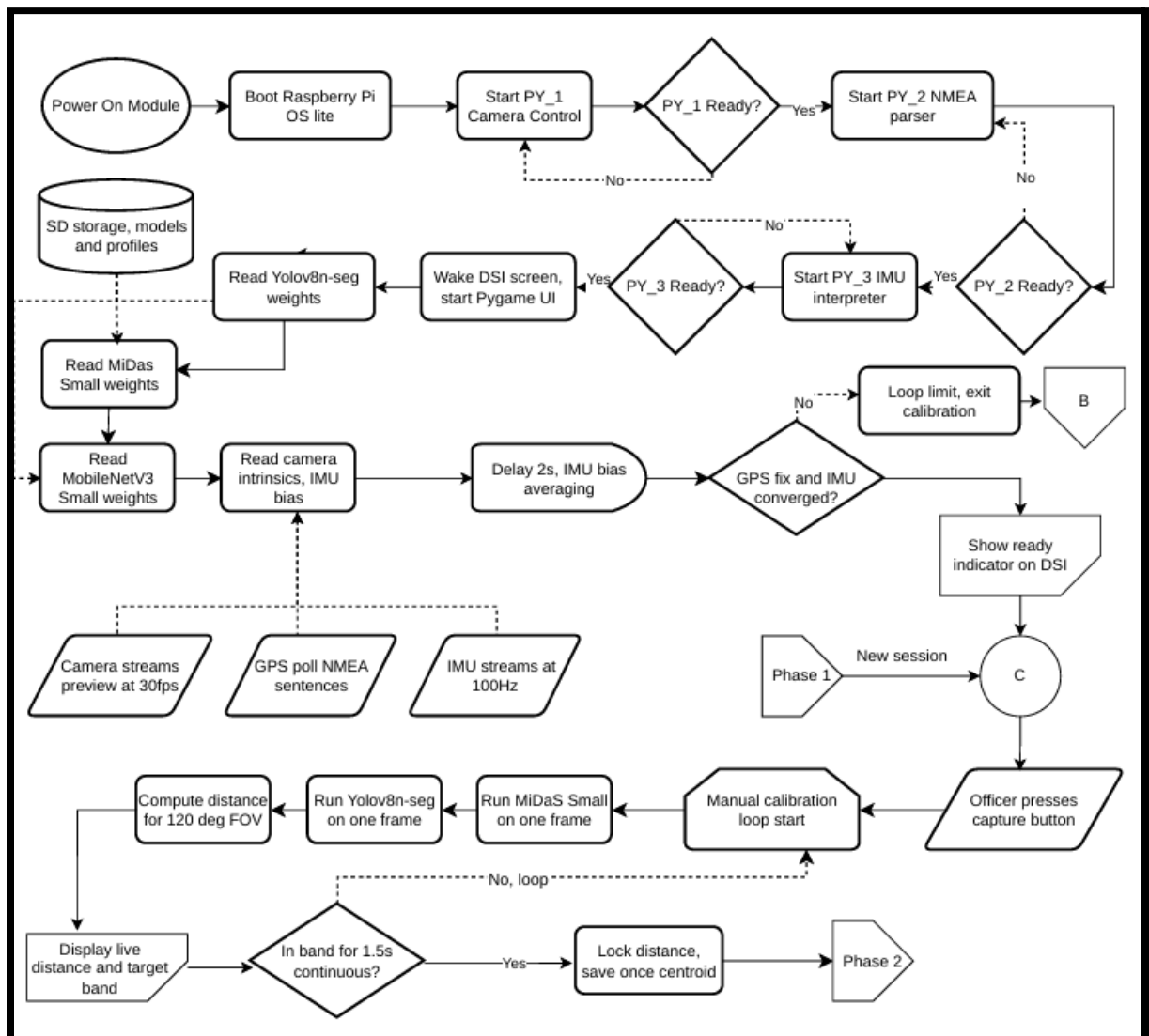


Figure 7. Phase 1 Boot Calibration Flowchart of an ATADS

As illustrated in figure 7, presents the Phase 1 Boot Calibration Flowchart of the Automated Traffic Accident Documentation System (ATADS), illustrating the sequential and conditional logic governing the system's initialization, model loading, sensor integration, and pre-capture calibration procedures prior to any accident documentation activity. The flowchart commences with a terminal symbol labeled "Power On Module," after which the system proceeds to boot the Raspberry Pi OS Lite environment — the lightweight operating system selected to minimize resource overhead on the constrained 2GB RAM platform. Following the OS boot, the system sequentially initiates three concurrent Python-based control processes: PY_1 (Camera Control), PY_2 (NMEA Parser for GPS), and PY_3 (IMU Interpreter), each governed by an individual readiness decision gate that enforces a retry loop until the respective process achieves a ready state, thereby ensuring no downstream operation is invoked on an uninitialized subsystem. Concurrently, the system reads three pre-trained lightweight model weight files — YOLOv8n-seg, MiDaS Small, and MobileNetV3 Small — from a dedicated SD storage component (represented by the stored data drum symbol), which are subsequently loaded into ONNX Runtime or TFLite inference engines for real-time deployment. Upon successful module initialization, the flowchart directs the system to wake the DSI touchscreen display and launch the Pygame user interface, after which camera intrinsics and IMU bias parameters are ingested to configure sensor-specific corrections. Three parallel sensor streams are then activated — camera preview at 30 frames per second, GPS NMEA sentence polling, and IMU data streaming at 100Hz — converging at a 2-second delay block that performs IMU bias averaging to reduce accelerometer and gyroscope drift errors. The calibration decision gate evaluates whether a valid GPS fix has been acquired and IMU values have converged within acceptable bounds; a

negative outcome triggers a dashed loop returning to the bias averaging stage, while repeated failures beyond a defined loop limit route the system to an off-page connector labeled B, which refers to the Exit Calibration Fallback sub-process detailed in Figure 6. Upon a positive calibration outcome, the system displays a readiness indicator on the DSI screen and routes to on-page connector C, where the officer initiates the capture sequence by pressing a designated button, triggering the manual calibration loop, parallel MiDaS and YOLOv8 inference on a single captured frame, distance computation across the 75-degree field of view, and a live-distance display; a final decision gate confirms whether the measured distance remains within the acceptable target band for a continuous 1.5 seconds before locking the distance and saving the object centroid, concluding Phase 1 and transitioning to Phase 2 of the ATADS pipeline via an off-page connector.

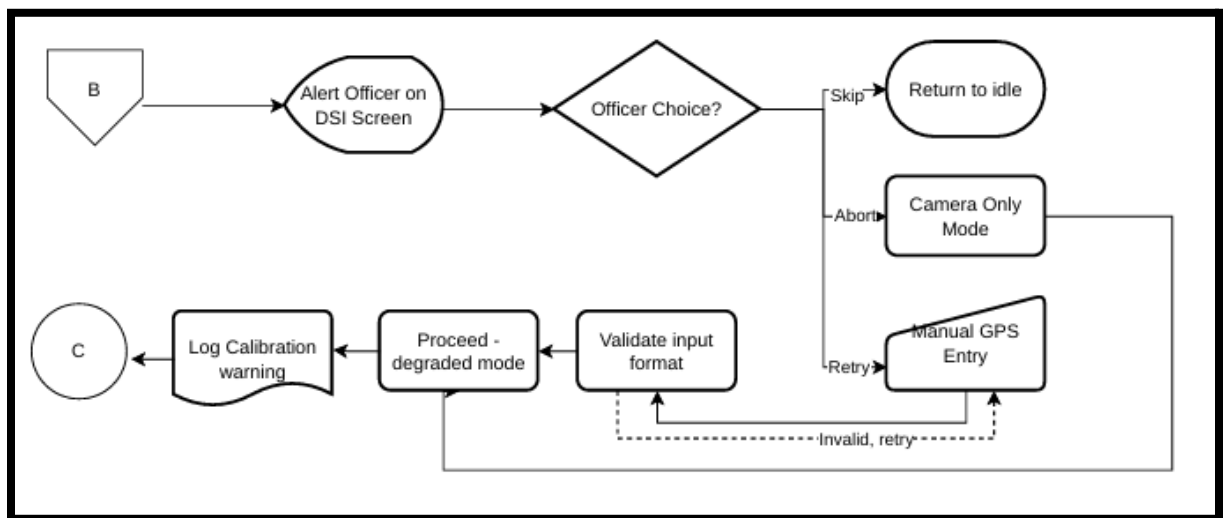


Figure 8. Phase 1 Exit Calibration FallBack Off-Page Connector

As illustrated in Figure 8, the diagram presents the **Exit Calibration Fallback sub-process** of the ATADS Phase 1 pipeline. This sub-process is structured as an

off-page connector sub-diagram and is entered only through **Connector B**. It is activated when the GPS and IMU calibration loop in Figure 7 reaches its defined iteration limit without achieving proper convergence. This prevents the system from continuing under conditions where geospatial or inertial data are not yet verified.

Upon entry at Connector B, the system immediately triggers an alert notification on the DSI touchscreen. This alert informs the attending officer that the calibration process has failed to converge. After this warning, the system presents a three-branch decision point labeled “**Officer Choice?**” This step helps ensure that the fallback action is not done automatically without the officer’s awareness. Instead, it requires deliberate human intervention before the system continues.

The first branch, labeled “**Skip,**” directs the system to the “**Return to Idle**” terminal symbol. This means that the current documentation attempt is stopped, and ATADS returns to its standby state. This is the most conservative option and is appropriate when the scene condition or hardware state does not allow reliable documentation to continue.

The second branch, labeled “**Abort,**” leads to the “**Camera Only Mode**” process block. In this mode, the system continues using only visual data from the camera subsystem while bypassing GPS- and IMU-dependent computations. This represents a degraded operating mode because the system can still capture visual evidence, but it cannot fully support geospatial positioning or orientation-based accuracy.

The third branch, labeled “**Retry,**” directs the officer to the “**Manual GPS Entry**” input block. This block allows the officer to manually enter geospatial

coordinates through the touchscreen keyboard. After the coordinates are entered, the system moves to the **“Validate Input Format”** decision gate, which checks whether the latitude and longitude values are structurally correct and reasonable.

If the entered GPS values are invalid, the system loops back to the manual entry block through a retry path. This allows the officer to correct the input before the process continues. This validation step is important because manually entered coordinates must still meet the required format before they can be accepted by the system.

Both the **Camera Only Mode** path and the successful **Manual GPS Entry** path converge at the **“Proceed — Degraded Mode”** process block. This indicates that the system will continue the documentation session, but under reduced data integrity. In this condition, the system acknowledges that some geospatial or inertial data may not be fully verified.

After proceeding in degraded mode, the system executes the **“Log Calibration Warning”** document process. This records the calibration failure event and attaches a warning flag to the output diagram package. This flag is important because it clearly indicates that the produced accident diagram contains unverified geospatial data, which must be considered during review, legal use, or forensic documentation.

Finally, the fallback sub-process ends by routing control back to **Connector C** in Figure 7. This returns the system to the main capture sequence at the officer-initiated capture trigger. Through this fallback design, ATADS prevents silent

calibration failure, preserves officer awareness, and maintains transparency in the documentation workflow.

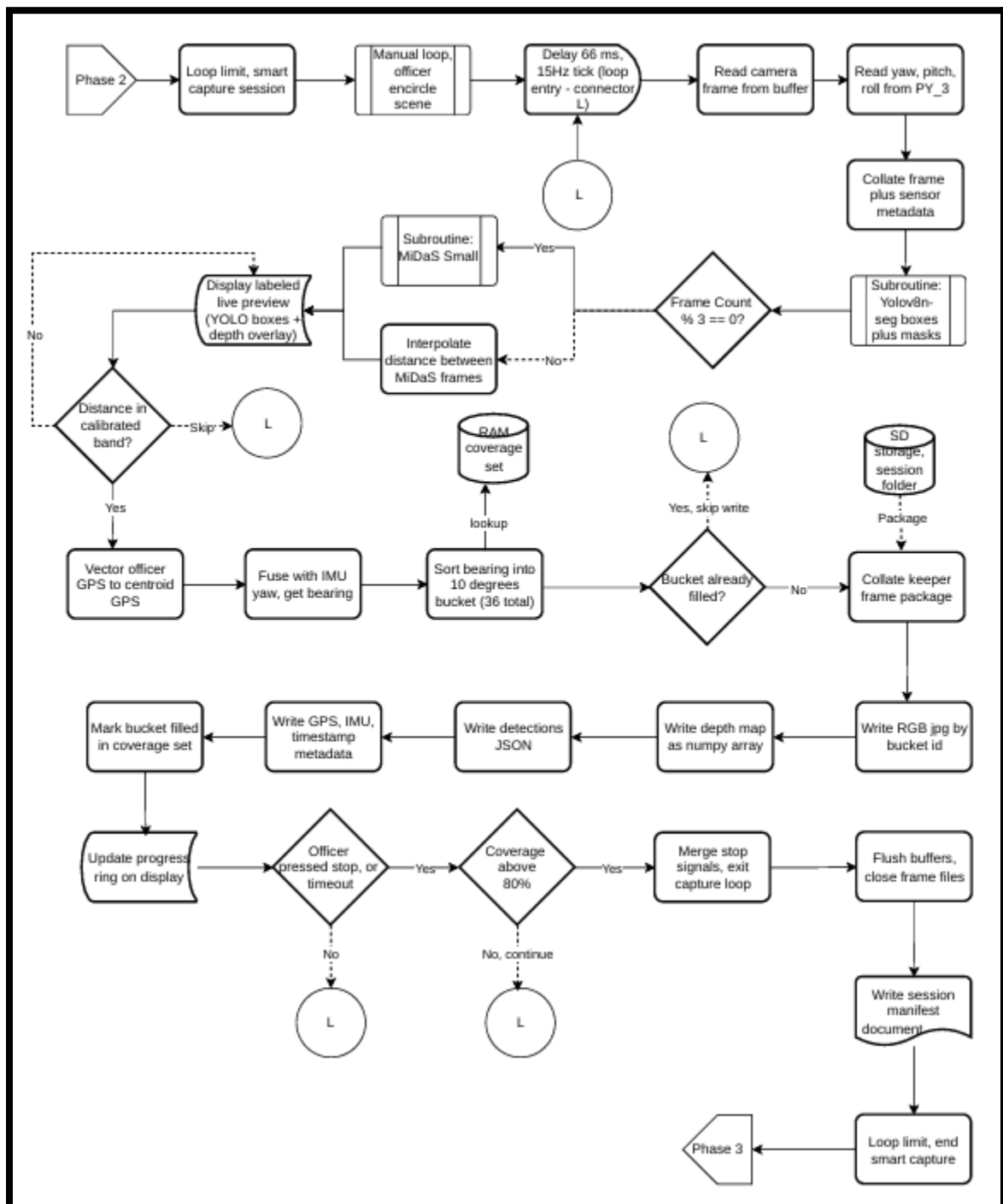


Figure 9. Phase 2 Smart Capture Session

Figure 9 illustrates the **Phase 2 Smart Capture Session** of the Automated Traffic Accident Documentation System (ATADS). This phase begins after the system exits Phase 1 through an off-page connector. It is immediately controlled by a loop-limit symbol, which represents the bounded smart capture session. This means that the capture activity follows a controlled cycle instead of running without limits.

The process then enters the **officer-controlled encirclement loop**. In this stage, the officer moves around or captures the accident scene from different directions while the system continuously collects visual and sensor data. This loop is important because ATADS needs multiple angles of the scene to support a more complete reconstruction.

A **66-millisecond delay** is applied in every cycle of the loop. This delay is based on the target capture rate of **15 frames per second**, which gives the system a steady rhythm for reading and processing frames. The on-page connector labeled **L** serves as the main loop re-entry point, allowing the process to return to the next capture tick after each cycle.

For every cycle, the system reads one camera frame from the buffer. At the same time, it retrieves inertial data from the **PY_3 sensor thread**, including yaw, pitch, and roll. These data are combined into one frame-sensor package, allowing the system to process the image together with the device's orientation readings.

The frame-sensor package is then submitted to the **YOLOv8n-seg predefined process subroutine**. This subroutine detects objects in the accident scene and returns bounding boxes and instance segmentation masks. These

outputs help identify scene elements such as vehicles, debris, obstructions, and other relevant objects.

A decision gate then checks whether the current frame count is divisible by three. This condition controls when the system will run the **MiDaS Small depth estimation** subroutine. On every third frame, the system performs full monocular depth inference to estimate depth from the camera image.

For frames that are not divisible by three, the system does not run full MiDaS inference. Instead, it interpolates distance values using the two most recent MiDaS outputs. This approach reduces the processing load on the Raspberry Pi 4 while still maintaining useful spatial estimation for the capture session.

After detection and depth estimation, the system displays the depth estimate and detection overlay on the touchscreen. This allows the officer to see what the system is detecting in real time. The visual feedback helps the officer monitor whether the scene is being captured properly.

The system then checks whether the computed object distance falls within the pre-calibrated measurement band. If the distance is not valid, the process returns to the display step with a warning annotation. This informs the officer that the current view may not be suitable for reliable capture.

If the officer chooses to override the invalid distance condition, the **Skip** path allows the system to bypass the write pipeline and return directly to connector **L**. This means the frame is not saved as a keeper frame, but the capture session continues. This option gives the officer control while still preventing poor-quality frame data from being stored.

When the distance validation is successful, the system computes a bearing vector from the officer's GPS coordinates to the detected object centroid. This bearing is then fused with the IMU yaw reading to produce a compass-referenced direction. This step helps connect detected objects to their directional position within the accident scene.

The computed bearing is then sorted into one of **thirty-six angular buckets**, with each bucket representing a **10-degree viewing angle**. These buckets are stored in a RAM-resident coverage set, represented by a stored-data symbol. This method helps the system track which scene directions have already been captured.

A second decision gate checks whether the assigned angular bucket already contains a keeper frame. If the bucket already has one, the current frame is discarded and the process returns to connector **L**. This prevents redundant data from being saved and helps ensure that the system focuses on collecting diverse viewing angles.

If the bucket is still empty, the system creates a **keeper frame package**. This package includes the RGB image, depth map, detection data, and GPS-IMU-timestamp metadata. These files are then written sequentially to the SD storage session folder.

The RGB image is saved using the bucket identifier, followed by the depth map stored as a NumPy array. The system also saves the detections in JSON format and records the GPS, IMU, and timestamp metadata. After these files are saved, the bucket is marked as filled in the coverage set.

The progress ring on the display is then updated. This gives the officer a visual indication of how much angular coverage has already been captured. The progress display helps the officer understand whether more scene angles are still needed.

Next, the system checks whether the officer has pressed the stop button or whether the session has reached the timeout threshold. If neither condition is met, the process loops back to connector **L** and continues to the next capture cycle. This keeps the smart capture session running until a valid stopping condition occurs.

If the stop button is pressed or the timeout threshold is reached, the system proceeds to a second coverage-adequacy check. This gate verifies whether the angular coverage exceeds **80 percent**. If coverage is not enough, the system forces the session to continue so that the accident scene can still be captured from enough directions.

When both stopping conditions are satisfied, the system merges the stop signals. It then flushes and closes the frame file handles to ensure that saved data are properly written to storage. This prevents incomplete or corrupted files from being used in later reconstruction.

After closing the files, the system writes a **session manifest** to disk. This manifest acts as a formal record of the capture session and contains the organized references to the saved data. It helps prepare the output for the next phase of processing.

Finally, the loop-limit symbol closes the smart capture session. Control is then transferred through an off-page connector to **Phase 3**, where the captured frames,

depth data, detections, and metadata can be used for scene reconstruction. This phase is important because it helps ensure that ATADS collects enough directional, visual, and sensor-supported data before generating the reconstructed accident scene.

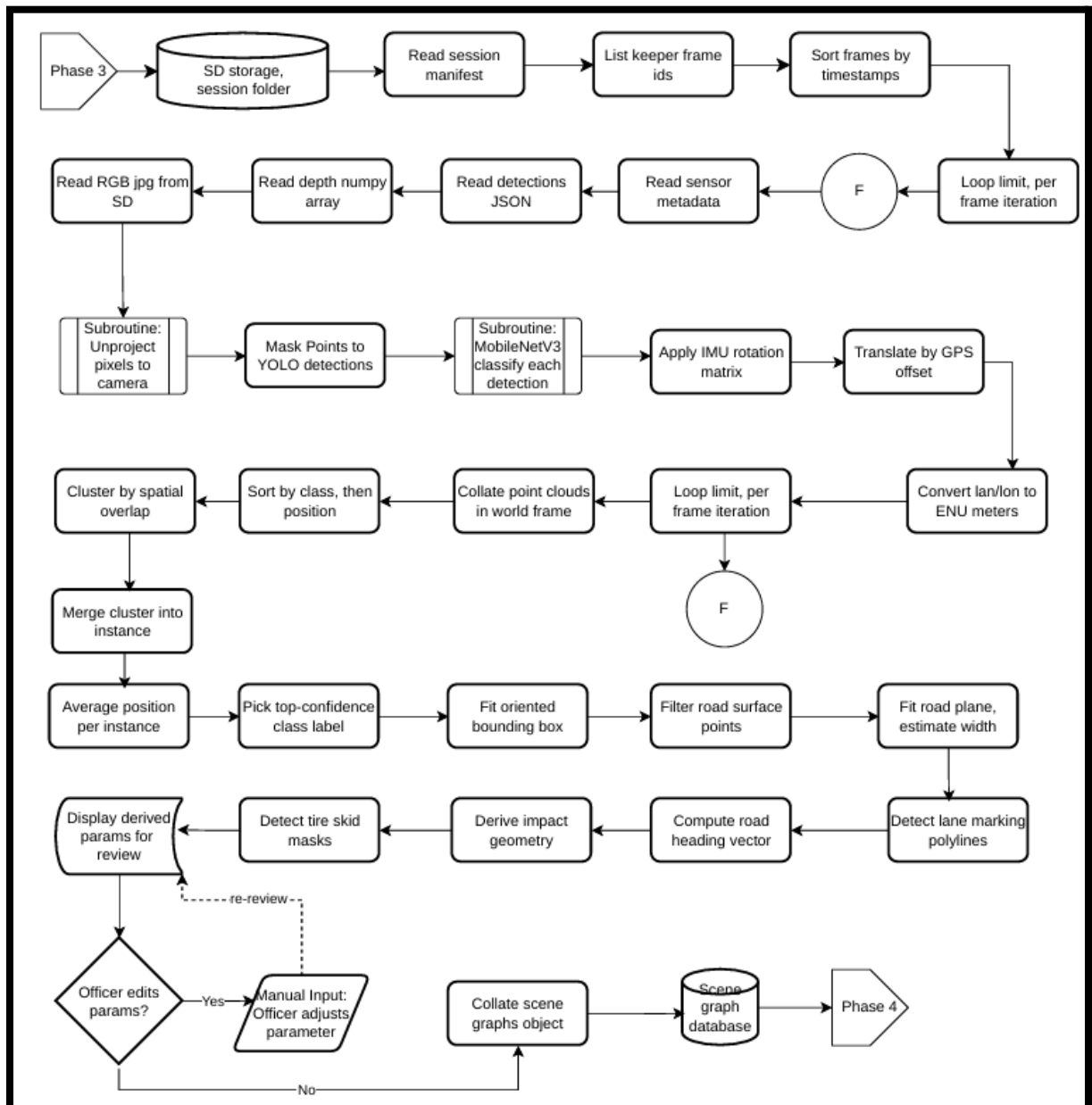


Figure 10. Phase 3 Scene Reconstruction from Multi-Frame Captures

Figure 10 presents the **Phase 3 Scene Reconstruction pipeline** of the Automated Traffic Accident Documentation System (ATADS). This phase begins after Phase 2 through an off-page connector. Its main purpose is to convert the saved smart capture data into a structured three-dimensional reconstruction of the accident scene.

The process first accesses the **SD storage session folder**, shown as a stored-data drum symbol. From this storage, the system reads the session manifest. The manifest is important because it lists and organizes the keeper frames collected during the smart capture session in their correct time order.

A loop-limit symbol then opens the **per-frame processing cycle**. This means that each saved frame is processed one at a time. The on-page connector labeled **F** marks the entry point of this repeated loop and allows the system to return to the next frame until all keeper frames have been processed.

For every frame, the system loads four important data assets. These are the RGB image from the SD card, the monocular depth estimate stored as a NumPy array, the YOLOv8n-seg detection records saved in JSON format, and the IMU and GPS metadata from the sensor logs. These data provide the visual, depth, object detection, and positioning information needed for reconstruction.

These four data streams then enter the **3D processing pipeline**. In this stage, the system reprojects each YOLO-detected pixel cluster from the flat image plane into camera-space three-dimensional coordinates. This is done using the stored camera intrinsic matrix, which helps the system understand how image pixels relate to real spatial positions.

After reprojection, the system produces a point cloud for the detected objects. The point cloud is then masked so that only the points inside the YOLO instance boundaries are retained. This prevents unrelated background points from being included in the object reconstruction.

The masked point clusters are then sent to **MobileNetV3-Small** for fine object classification. This process assigns a semantic class label to each detected object. In simpler terms, the system does not only know that an object exists; it also identifies what kind of object it may be, such as a vehicle-related element, road evidence, or other accident-scene feature.

Next, the classified point cloud is registered into a **global coordinate frame**. This step allows data from different viewpoints to be placed into one shared scene space. Without this process, each frame would only represent the scene from the officer's camera position and would not align correctly with the other captured frames.

The registration process uses several transformations. First, the IMU rotation matrix corrects the camera's tilt and roll at the moment of capture. Second, the GPS offset moves the points from the camera-local position into a geographic reference. Third, the system converts geodetic latitude and longitude into **East-North-Up metric space**, making the data comparable across all frames.

Once every frame has been processed, the closing loop-limit symbol ends the per-frame iteration. The dashed loop-back path through connector **F** shows that the same process repeats for each frame until all sorted keeper frames are completed.

After the final frame, the system exits the loop and moves to post-loop reconstruction.

In the **post-loop instance reconstruction stage**, the point clouds from all frames are combined into one world-frame representation. The system sorts the points by semantic class and spatial position. This helps determine which detections from different angles refer to the same physical object in the accident scene.

A spatial overlap clustering process then groups related measurements together. For example, if the same vehicle appears in several frames from different directions, the system combines those measurements into one object instance. This improves the completeness and accuracy of the reconstructed scene.

Each merged object is then described using important properties. These include an averaged centroid position, the most confident semantic class label from MobileNetV3, and an oriented bounding box. The bounding box represents the object's size, direction, and spatial extent within the accident scene.

After object reconstruction, the system performs **road surface analysis**. Points not associated with detected objects are filtered to isolate the road surface. This is important because the road itself provides the reference for lane width, slope, direction, and accident geometry.

The system then fits a least-squares plane to the ground-level points. This helps estimate the carriageway width and road gradient. Lane marking polylines are also extracted from the RGB images to help define the road structure more clearly.

A road heading vector is then computed from the fitted plane normal. This vector establishes the main direction of travel or the principal road axis. It is important because accident interpretation often depends on how vehicles are positioned relative to the road direction.

The system then derives **impact geometry** by analyzing the angular and spatial relationship between vehicle instances and the road axis. This helps estimate how the vehicles may have approached or collided. Tire skid marks are also detected from the road surface image to provide possible evidence of braking before impact.

All derived geometric parameters are then displayed on the touchscreen for officer verification. This display step allows the officer to review the reconstructed scene before it is accepted. The officer can check whether vehicle positions, road direction, impact relationship, and evidence indicators appear reasonable.

A decision gate then checks whether the officer wants to adjust any parameter. If adjustment is needed, the process moves to a manual input step where the officer can edit the parameters. After editing, the system loops back to the display for another review, ensuring that corrections are checked again before acceptance.

If the officer accepts the reconstructed parameters, the system collates all verified scene details into a **scene graph object**. This scene graph contains the structured representation of the accident scene, including objects, positions, classifications, geometry, and evidence indicators.

The verified scene graph is then written to the **scene graph database**, shown as a stored-data drum symbol. This storage step is important because it preserves the reconstructed accident scene for the next phase of the system.

Finally, an off-page connector transfers control to **Phase 4**. At this point, ATADS carries forward a fully attributed, georeferenced, multi-instance three-dimensional reconstruction of the traffic accident scene. This reconstructed scene becomes the basis for rendering the final accident diagram and generating the official documentation output.

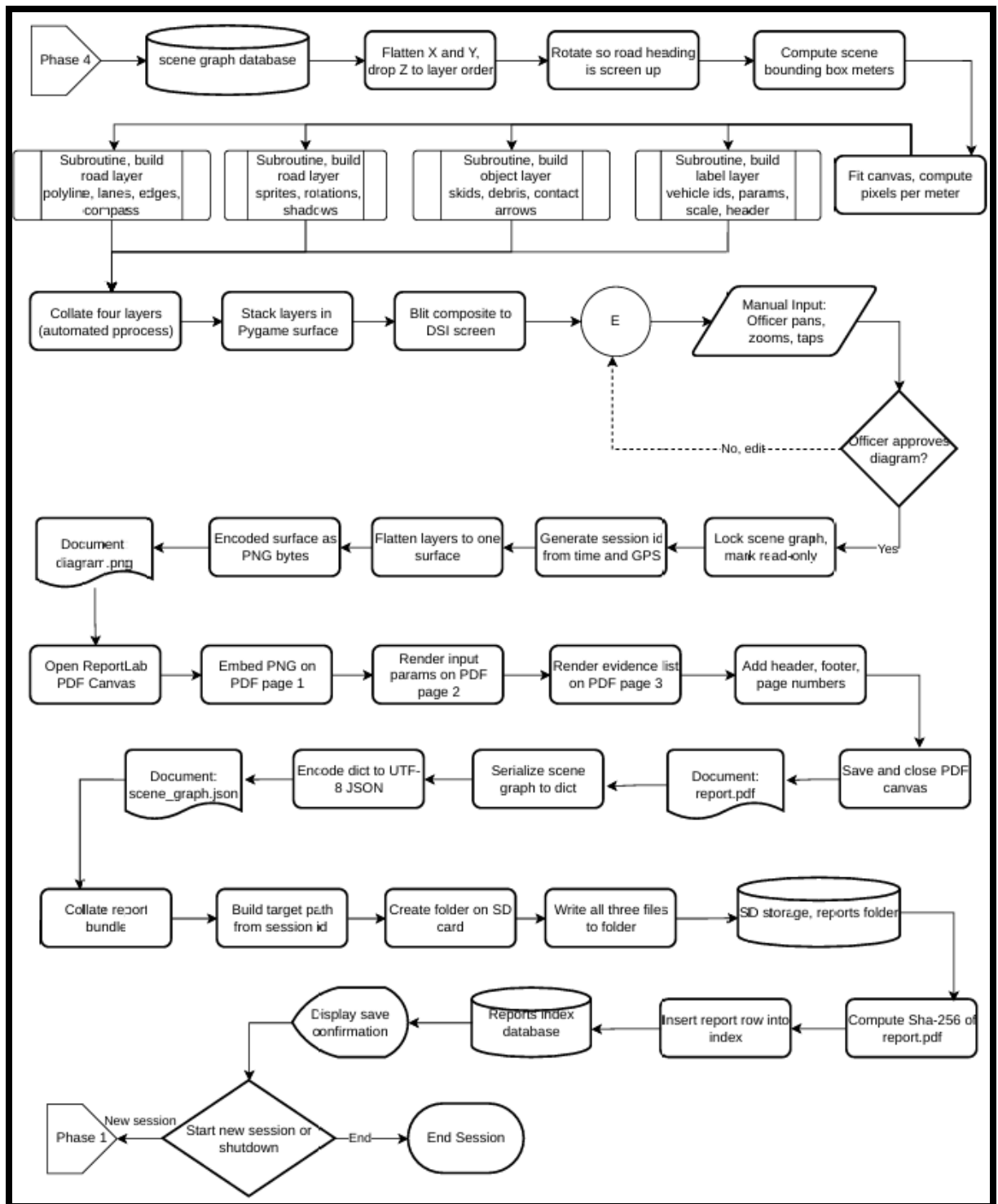


Figure 11. Phase 4 Diagram Rendering and Report Generation

Figure 11 illustrates the **Phase 4 Diagram Rendering and Report Generation pipeline** of the Automated Traffic Accident Documentation

System (ATADS). This phase serves as the final computational stage of the system, where the verified scene graph from Phase 3 is converted into a finished accident diagram and report package. It is important because this stage transforms processed scene data into outputs that can be reviewed, saved, and used as formal documentation.

The phase begins through an off-page connector from Phase 3. The system immediately retrieves the verified scene graph from the database, shown as a stored-data drum symbol. This scene graph contains the reconstructed accident scene, including object positions, road layout, vehicle details, evidence markers, and related geometric information.

After retrieving the scene graph, the system performs three coordinate normalization steps. First, the three-dimensional point cloud is flattened into a two-dimensional plane by using the X and Y coordinates for the diagram layout, while the Z-axis values are reassigned as drawing layers. This allows elevation-related information to still influence how the diagram is arranged without making the final output difficult to read.

Second, the system rotates the entire scene so that the computed road heading vector aligns with the vertical screen axis. This is important because accident diagrams must be oriented in a clear and consistent way. By aligning the road direction properly, the final diagram becomes easier for officers, reviewers, and other readers to understand.

Third, the system computes a metric bounding box around the scene. This bounding box determines how the diagram will fit on the screen and in the report. It helps control scaling so that objects, distances, and road features are represented in a more accurate and organized manner.

The normalized scene data is then passed into four parallel rendering processes. These are the road layer, road sprite layer, object layer, and label layer. Each layer focuses on a different part of the accident diagram, allowing the system to build the final output in an organized and readable format.

The **road layer** includes road polylines, lane markings, road edges, and a compass indicator. This layer provides the base structure of the accident scene. It helps readers understand the road layout, direction, and driving environment where the collision occurred.

The **road sprite layer** contains vehicle silhouettes, rotation transforms, and drop shadows. This layer represents the vehicles involved in the accident and places them according to their reconstructed positions and headings. This is important because vehicle orientation is one of the key details needed in accident scene interpretation.

The **object layer** includes tire skid marks, debris fields, and contact point arrows. These objects help show possible evidence of braking, impact location, and collision movement. This layer supports the evidentiary value of the diagram because it includes accident-specific physical indicators.

The **label layer** adds vehicle identifiers, measured parameters, scale bar, and report header. This makes the diagram easier to interpret because important details are directly labeled. The scale bar is especially important because it helps maintain a sense of measurement and proportion in the final accident diagram.

A separate process computes the **pixel-per-meter ratio** by fitting the canvas size to the metric bounding box. This step helps ensure that the diagram is not only visually arranged but also metrically consistent. It helps prevent the scale distortion problem commonly found in manual accident sketches.

After the layers are created, the system collates and composites them using the Pygame surface rendering engine. The layers are stacked into one complete accident diagram and displayed on the DSI touchscreen. This allows the officer to inspect the rendered diagram before it becomes finalized.

The on-page connector labeled **E** serves as the re-entry point for the officer interaction loop. If the officer chooses to edit the diagram, the process returns to this point. This loop gives the officer a chance to review the diagram carefully instead of accepting the automated output immediately.

The officer can interact with the diagram through the touchscreen by panning, zooming, and tapping scene elements. These interactions allow the officer to adjust attributes when necessary. This is important because the

system still preserves human verification and does not rely only on automated rendering.

A decision gate then checks whether the officer approves the rendered diagram. If the officer selects “**No, edit,**” the system returns to connector **E** and allows further review and correction. This helps ensure that errors or unclear diagram elements can be corrected before finalization.

If the officer approves the diagram, the system locks the scene graph and marks it as read-only. This prevents further modification after approval. This step is important for maintaining the integrity of the final accident record and protecting the diagram from accidental changes.

After approval, the system generates a unique session identifier using the GPS timestamp and geographic coordinates. This identifier provides a distinct reference for the report package. It helps connect the final output to the specific accident scene and capture session.

The rendered Pygame surface layers are then flattened into a single composite image. This image is encoded as PNG bytes and saved as **diagram.png**. This file serves as the visual accident diagram that can be viewed separately or embedded into the final report.

The system then begins the PDF generation process. A ReportLab canvas is opened, and the PNG accident diagram is embedded on the first

page. This gives the report a clear visual summary of the reconstructed accident scene.

On the second page, the system places the measured input parameters and officer-verified scene attributes. These may include important accident details, confirmed measurements, and verified descriptions. This page supports the diagram by explaining the documented scene information in text form.

On the third page, the system adds the photographic evidence list with GPS-tagged capture metadata. This connects the report to the actual captured frames from the smart capture session. It strengthens traceability because the visual evidence is linked to its location and capture information.

After the report pages are prepared, the system applies a unified header, footer, and sequential page numbers. The canvas is then saved and closed, producing the completed **report.pdf** document. This PDF becomes the main human-readable documentation output of ATADS.

In parallel, the scene graph object is converted into a Python dictionary, encoded as UTF-8 JSON, and saved as **scene_graph.json**. This file provides a machine-readable version of the reconstructed accident scene. It is useful for future review, system processing, or technical validation.

The system then collates **diagram.png**, **report.pdf**, and **scene_graph.json** into one report bundle. These three files work together:

the PNG provides the visual diagram, the PDF provides the formal report, and the JSON preserves the structured scene data. This makes the output both readable for people and usable for the system.

A target folder path is created using the session identifier. The system then creates a session folder on the SD card inside the reports directory. All three files are written into this folder, ensuring that the outputs are stored together under one organized report package.

After saving the files, the system computes a **SHA-256 cryptographic hash** of the report.pdf. This hash acts as a tamper-evident integrity fingerprint. It helps show whether the report has been changed after generation.

The system also inserts an index row into the reports index database. This row contains the session ID, file path, timestamp, and hash value. This supports future retrieval and chain-of-custody verification because each report can be traced through its stored reference and integrity marker.

A save confirmation message is then displayed on the DSI touchscreen. This informs the officer that the report package has been successfully saved. This confirmation is important because the user must know that the final output is already stored before ending or starting another session.

Finally, the system asks whether the officer wants to start a **New Session** or **End** the process. If **New Session** is selected, the system returns to Phase 1 at the capture entry point while keeping active sensor threads

available. If **End** is selected, the process moves to the terminal symbol labeled **End Session**, formally closing the ATADS workflow.

Entity Relationship Diagram

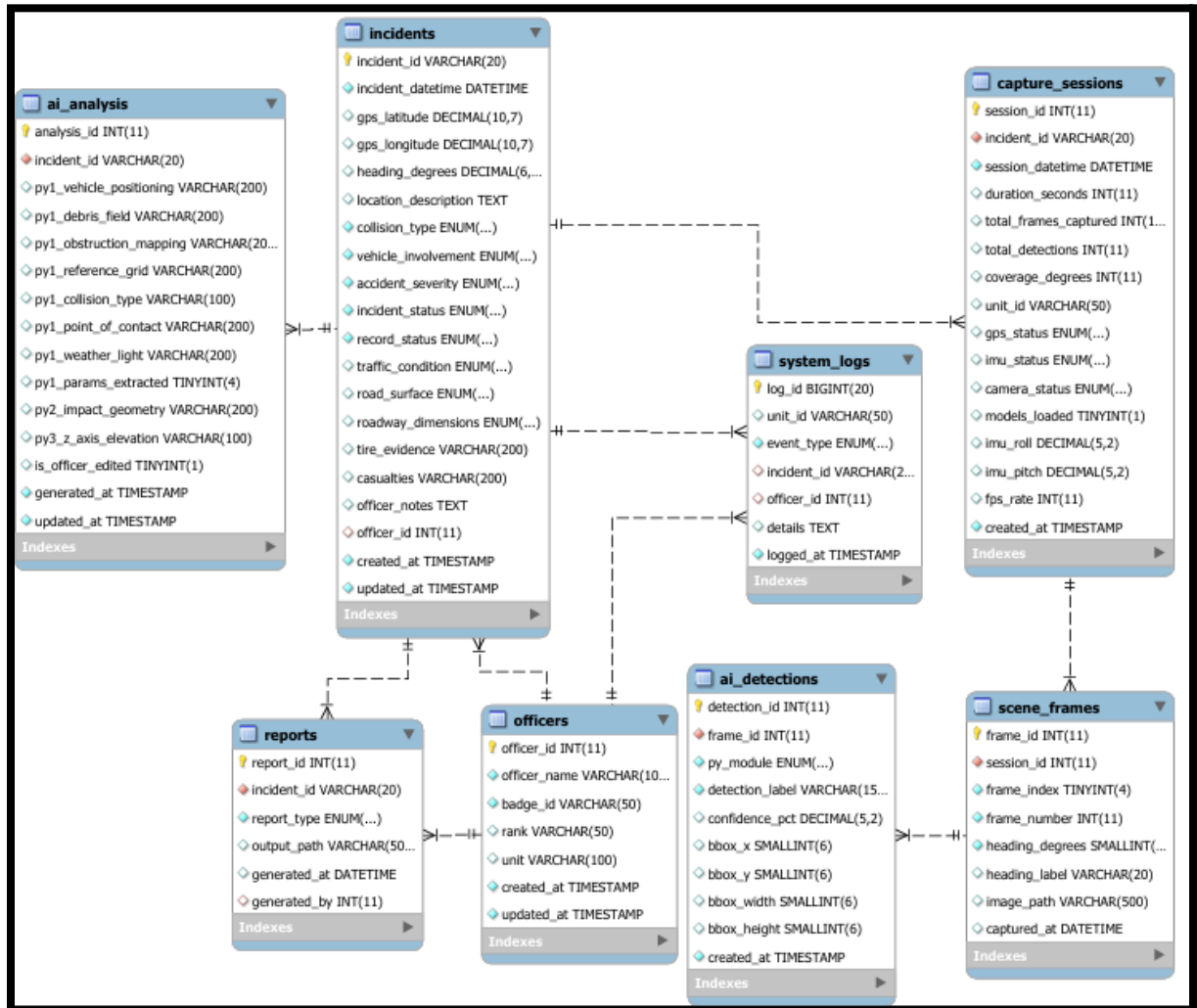


Figure 12. ERD of ATADS Application

As illustrated in figure 12, the proposed **Entity Relationship Diagram (ERD)** of the **Automated Traffic Accident Documentation System (ATADS)** shows how the system stores, connects, and manages all data needed for traffic accident documentation. It was designed to support the full process of recording an accident,

capturing scene evidence, analyzing visual and sensor data, generating reports, and monitoring system activity. Through this structure, ATADS can maintain organized, accurate, and traceable records that help reduce common problems in manual documentation, such as missing measurements, unclear sketches, inconsistent details, and unreliable scene interpretation.

The **officers table** stores the information of authorized personnel who operate and manage accident records in the system. It includes details such as the officer's name, badge number, rank, and assigned unit. This entity is important because every incident must be connected to the responsible officer, ensuring accountability and proper identification of the person who handled the documentation.

The **incidents table** is the central entity of the ERD because it contains the main information about each traffic accident. It stores the incident ID, date and time, GPS coordinates, location description, collision type, vehicle involvement, accident severity, incident status, traffic condition, road surface, roadway dimensions, tire evidence, casualties, and officer notes. Since the other records are connected to a specific incident, this table serves as the main reference point for the entire documentation process.

The **capture_sessions table** records each actual scene documentation session performed by the ATADS device. It stores the session date and time, number of captured frames, number of detections, coverage degree, unit ID, GPS status, IMU status, camera status, model loading status, roll and pitch values, and frame rate. This table helps verify the condition and performance of the device

during evidence collection, making it useful for checking whether the captured data were complete and reliable.

The **scene_frames table** stores the individual images captured during a capture session. Each frame is linked to a specific session and includes information such as frame index, frame number, heading degree, heading label, image path, and capture time. These frames represent the different views of the accident scene and provide the visual basis for documentation, review, AI detection, and diagram generation.

The **ai_detections table** stores the results produced from the captured scene frames. It includes the detected label, AI module used, confidence percentage, and bounding box coordinates. This entity allows the system to record accident-related visual findings such as vehicle position, rollover evidence, debris, collision indicators, and other objects detected in the scene. By storing these detection results, the system can support more consistent and evidence-based accident analysis.

The **ai_analysis table** stores the summarized interpretation of the accident scene after the AI detection process. It includes structured information such as vehicle positioning, debris field, obstruction mapping, reference grid, collision type, point of contact, weather and lighting condition, impact geometry, and elevation data. This table is important because it converts raw detection results into more understandable investigation details that can assist the officer in reviewing and completing the accident record.

The **reports table** stores the final outputs generated by the system, such as PDF reports, 2.5D diagrams, or combined report packages. It includes the report

type, file path, generation date, related incident, and the officer who generated the report. This entity supports proper filing and retrieval of official accident documentation, allowing completed records to be reviewed, exported, and used for future reference.

The **system_logs table** records important system activities and events, such as system startup, GPS lock, AI model loading, capture start, capture stop, record creation, record updates, report generation, and system errors. This table provides an activity trail that helps monitor the system's reliability and supports troubleshooting during testing and actual use.

Overall, the ERD provides a structured data foundation for ATADS by connecting officers, incidents, capture sessions, scene frames, AI detections, AI analysis, reports, and system logs. Each table has a specific role, but together they support a complete and standardized accident documentation workflow. This design improves traceability, reduces documentation errors, strengthens data organization, and helps produce clearer traffic collision records compared to manual sketch-based methods.

Prototype Design

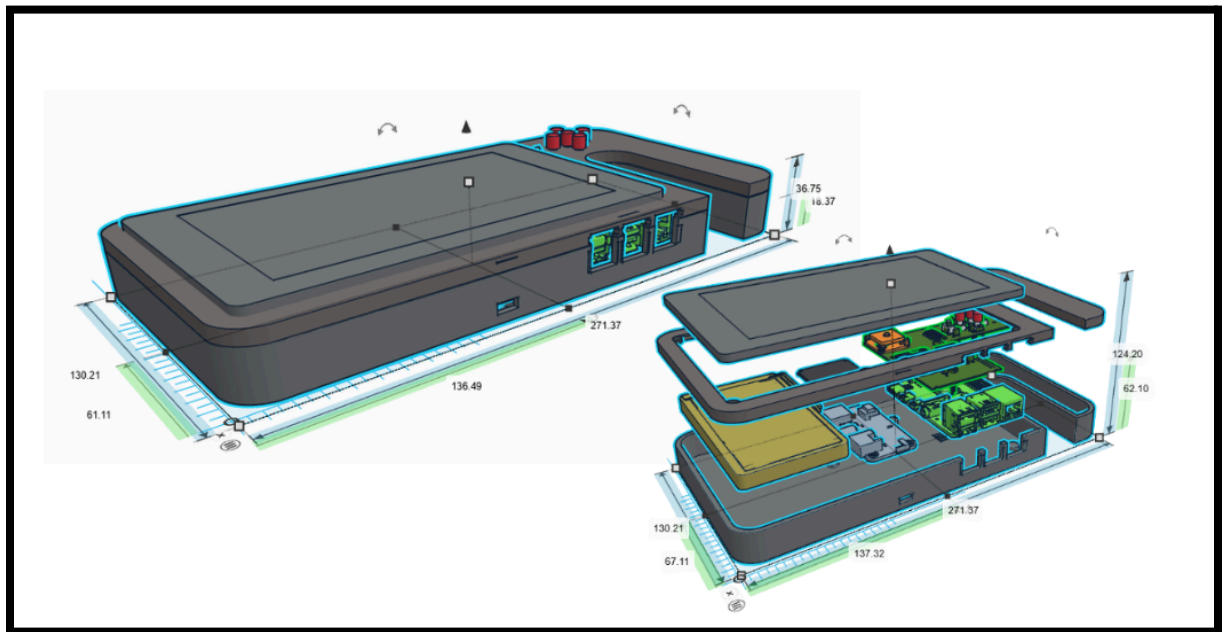


Figure 13. ATADS's Breakout

The ATADS (Automated Traffic Accident Documentation System) prototype, as illustrated in the figure 13, is a compact, ruggedized enclosure unit measuring 130.21 mm (length) × 136.49 mm (width) × 36.75 mm (height), designed to house all the electronic and sensing components necessary for automated traffic accident documentation. The exploded view reveals a multi-layered internal architecture consisting of integrated printed circuit boards (PCBs) for processing and communication, dedicated sensor modules (shown in green), a power/energy storage component (shown in gold/yellow), and supplementary electronic subsystems — all neatly enclosed within a hardened, weather-resistant shell featuring ventilation and connectivity ports along its side panel, as seen in the assembled view on the left. Despite its remarkably slim and compact form factor of only 36.75 mm in height, the system's modular internal layout helps ensure ease of maintenance and component replacement, while the protective outer casing with its

raised perimeter border and secure base footprint helps ensure stability and durability in real-world roadside deployment, enabling the ATADS unit to autonomously capture, process, and document traffic incident data without requiring manual intervention.

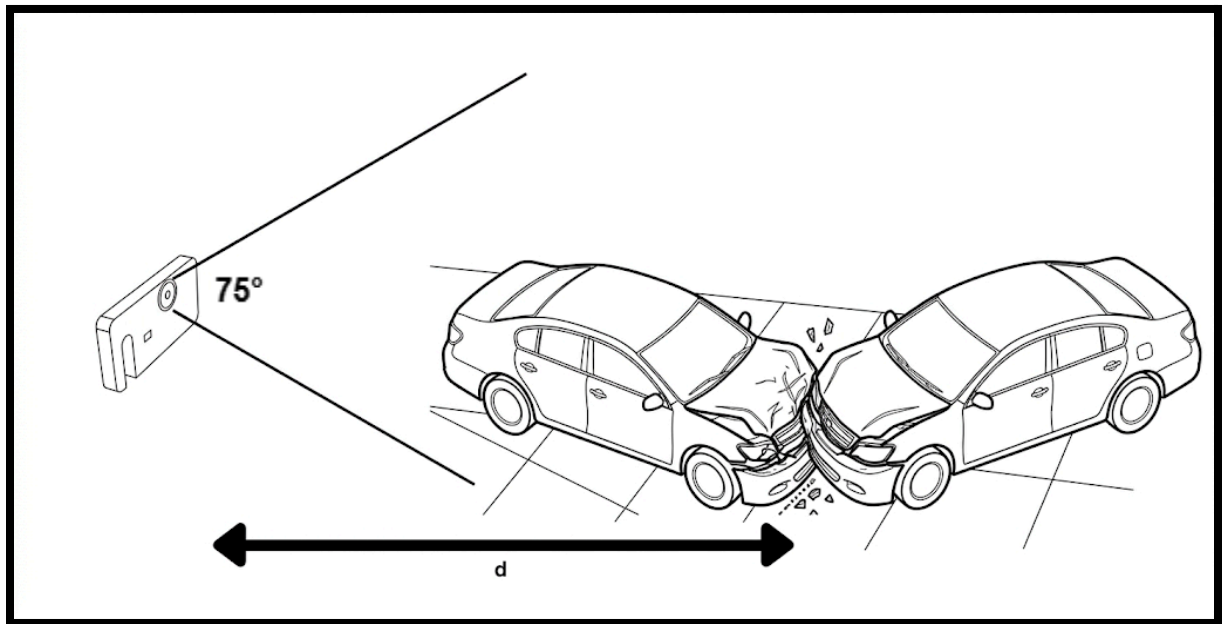


Figure 14. ATADS distance measurement

In Figure 14, the selected camera has a **75° field of view**. Field of view refers to the total angle that the camera can capture from its lens. A wider angle allows the camera to capture a larger scene, while a narrower angle requires the camera to be placed farther from the target area.

For this study, the camera is used to capture a possible road accident scene. The main purpose of the field-of-view analysis is to determine how far the camera should be placed from the accident area so that the full scene can be seen clearly.

Since the camera view spreads equally to the left and right side of its center line, the total field of view is divided into two equal parts.

$$\frac{75^{\circ}}{2} = 37.5^{\circ}$$

This means that each side of the camera's view covers **37.5°** from the center line. The half-angle is used because the camera's viewing area can be divided into two right-angle triangles. One triangle represents the left side of the view, and the other triangle represents the right side. Using right-angle triangles makes it possible to apply the tangent function in the computation.

The formula used to compute the required camera distance is shown in Equation 1.

$$d = \frac{W}{2 \times \tan(\theta/2)}$$

(Eq.1)

Where:

Symbol	Meaning
<i>d</i>	Required distance from the camera to the scene
<i>W</i>	Total width of the scene to be captured
<i>θ</i>	Total field of view of the camera
(<i>θ</i> /2)	Half of the camera's field of view

For a **75° camera**, the half-angle is **37.5°**. Therefore, the formula becomes:

$$d = \frac{W}{2 \times \tan(37.5^\circ)}$$

The tangent of **37.5°** is approximately:

$$\tan (37.5) \approx 0.767326988$$

Substituting this value into the formula gives:

$$d = \frac{W}{2 \times 0.767}$$
$$d = \frac{W}{1.534}$$

This equation shows that the required camera distance depends on the total width of the scene. If the scene is wider, the camera must be placed farther away.

Table 7. The estimated width of a typical accident scene.

Scene Element	Approx. Width(W)
Single Car width	~1.8 – 2.0 m
Two collided cars (side by side)	~4.0 – 5.0 m
With road buffer/margins	~6.0 – 8.0 m

For table 7, the accident scene is estimated to be 6 meters to 8 meters wide. This estimate includes the vehicles and a small road allowance around the accident area.

For a scene width of 6 meters, the distance is computed as follows:

$$d = \frac{6}{1.534}$$
$$d \approx 3.91 \text{ m}$$

For a wider accident scene of 8 meters, the distance is computed as follows:

$$d = \frac{8}{1.534}$$
$$d \approx 5.21 \text{ m}$$

This means that the camera must be placed approximately 5.21 meters away to capture an 8-meter-wide accident scene.

Based on the results, a 75° camera should be positioned approximately 3.9 meters to 5.2 meters away from the accident scene. This distance range allows the camera to capture a typical two-vehicle accident area, including a small allowance around the vehicles. The result also shows that a 75° field of view provides a more focused viewing area compared with wider-angle cameras. However, because the angle is narrower, the camera needs to be placed farther from the scene to capture the full accident area. In actual installation, the camera should not be placed at the exact computed distance only. A small allowance should be added because road conditions, camera height, mounting angle, and actual vehicle positions may affect the captured view.

Therefore, a practical camera placement distance of about 4 meters to 5.5 meters is recommended. The field-of-view analysis shows that a 75° camera can

capture a typical accident scene when placed at the correct distance. For a scene width of 6 meters, the required distance is approximately 3.91 meters. For a wider scene of 8 meters, the required distance is approximately 5.21 meters. Therefore, the selected camera is suitable for accident monitoring when installed at an estimated distance of 4 meters to 5.5 meters from the target scene. This placement helps ensure that the full accident area remains within the camera frame.

Wire Frames



Figure 15.1. ATADS Welcome and System Startup Screen

Figure 15.1. ATADS Welcome and System Startup Screen shows the opening screen of the **Automated Traffic Accident Documentation System (ATADS)**. It displays the system name, ATADS logo, location context, and the instruction “**Tap Anywhere to Continue.**” The screen also shows the current system condition as

System Offline, which informs the user that the device has not yet fully connected or activated its required services.

This screen is important because it serves as the first interaction point between the investigator and the ATADS prototype. Before any accident record can be created or reviewed, the user must first enter the system through this interface. The simple layout helps reduce confusion, especially for field users who may be working under pressure during traffic accident response.

The design supports the goal of ATADS by providing a clean and direct startup interface. Since traffic investigators often work in urgent and stressful environments, the system must avoid unnecessary buttons or complicated instructions. This screen prepares the user for a guided workflow where the documentation process begins only after the system is opened and ready for use.

ATADS Incident Records						Q Search records... 10.3157N 123.8854E	
42 TOTAL		7 IN REVIEW		3 DRAFTS		32 COMPLETE	
Scene ID	Date / Time	Location	Collision Type	Vehicle Involvement	Status		
ACC_0911	Dec 14 2023 14:22	N. Bacalso Ave	Rollover	Car - Truck	COMPLETE	View	
ACC_0910	Dec 14 2023 09:45	M.J. Cuenco Ave	Side Impact	Motorcycle - Car	IN REVIEW	View	
ACC_0909	Dec 13 2023 16:10	Mandaue Bridge	Rear-End	Motorcycle - Car	COMPLETE	View	
ACC_0908	Dec 13 2023 11:30	Cebu South Rd	Multi-Vehicle	Multi-Vehicle (3+)	DRAFT	View	
ACC_0907	Dec 12 2023 08:15	Osmena Blvd	Hit and Run	Motorcycle - Motorcycle	CLOSED	View	
ACC_0906	Dec 12 2023 19:48	Colon St	T-Bone	Car - Truck	IN REVIEW	View	
ACC_0905	Dec 11 2023 13:05	V. Rama Ave	Pedestrian Strike	Vehicle - Pedestrian	COMPLETE		

CAPTURE

Start new recording

UNIT STATUS

GPS

IMU

Camera

Models

• LOCKED

• READY

• STANDBY

• LOADED

TODAY - DEC 14, 2023

Incidents

Weather

2

Clear / Bright

PNP/TARAS Standard

System Logs

ATADS-UNIT-01 | 7 of 42 records | 10.3157N 123.8854E

ONLINE

Figure 15.2. ATADS Incident Records Dashboard

ATADS Incident Records Dashboard from figure 15.2 presents the main **Incident Records Dashboard** of ATADS. It shows a summary of accident records, including the total number of cases, records under review, drafts, and completed reports. The dashboard also displays a table containing important incident details such as scene ID, date and time, location, collision type, vehicle involvement, and current status.

The right side of the dashboard contains the **Capture** button, which allows the officer to start a new accident scene recording. It also includes a **Unit Status** panel showing the condition of the GPS, IMU, camera, and AI models. These indicators are important because ATADS depends on sensor and AI readiness before capturing reliable accident data.

This dashboard from figure 15.2 is relevant to the prototype because it organizes all accident records in one place. Instead of searching through paper files or separate photo folders, the investigator can immediately view, search, and monitor the progress of each case. The status labels such as **Complete**, **In Review**, **Draft**, and **Closed** help users quickly understand which records still need action.

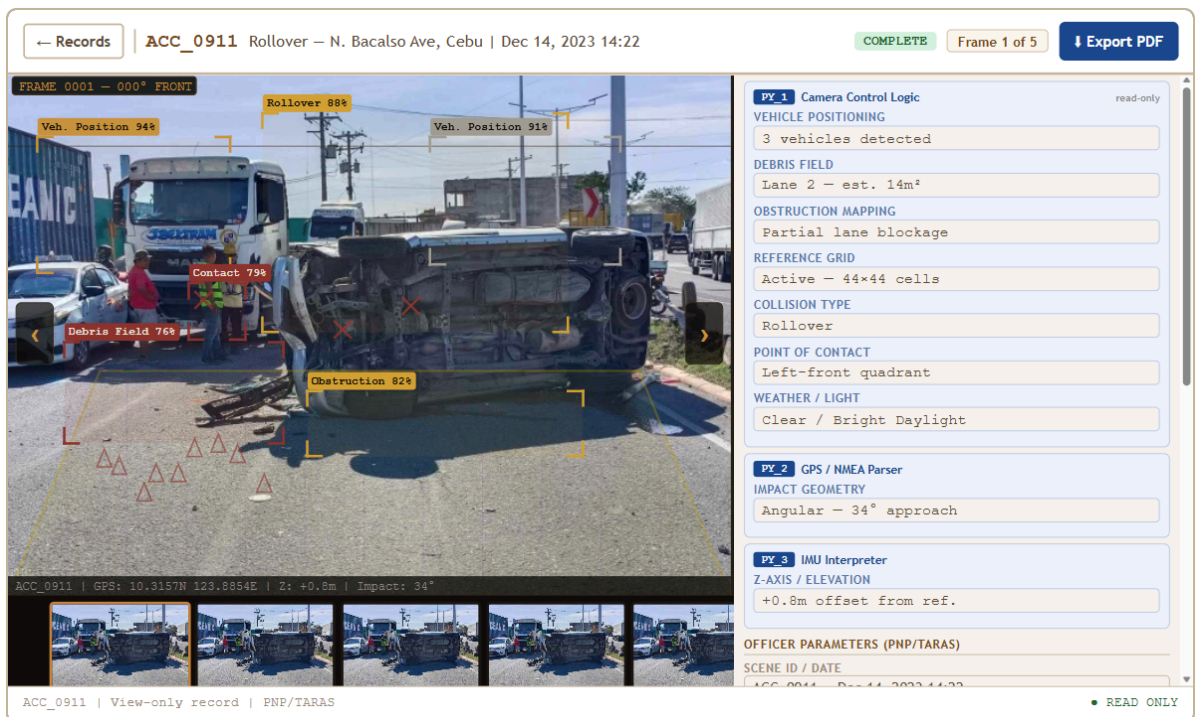


Figure 15.3. Completed Incident Record View with AI Analysis Panel

Figure 15.3 shows a completed accident record for **ACC_0911**. The left side displays the captured accident scene image with AI detection overlays, including vehicle position, rollover detection, contact point, debris field, and obstruction area. The bottom part shows multiple captured frames, allowing the officer to review the scene from different angles.

The right panel shows the AI-generated analysis grouped by system modules. **PY_1 Camera Control Logic** identifies visual accident elements such as vehicle positioning, debris field, obstruction mapping, reference grid, collision type, point of contact, and weather or lighting condition. **PY_2 GPS/NMEA Parser** provides impact geometry, while **PY_3 IMU Interpreter** provides elevation-related information.

Completed Incident Record View with AI Analysis Panel demonstrates how ATADS converts captured images into structured investigation data. The screen is

marked as **read-only**, which means the record is already completed and protected from further changes. This is important for maintaining the integrity of accident records, especially when reports may be used for official review, investigation, or legal documentation.

Records | ACC_0911 Rollover — N. Bacalso Ave, Cebu | Dec 14, 2023 14:22

COMPLETE Frame 1 of 5 Export PDF

FRAME 0001 — 000° FRONT

Veh. Position 94% Rollover 88% Veh. Position 91% Contact 73% Debris Field 76% Obstruction 82%

ACC_0911 | GPS: 10.9157N 123.8854E | Z: +0.8m | Impact: 34°

ACC_0911 | View-only record | PNP/TARAS

SCENE ID / DATE
ACC_0911 — Dec 14, 2023 14:22

LOCATION
N. Bacalso Ave, Cebu City

ACCIDENT SEVERITY (ART. 365 RPC)
☒ Injury ☐ Fatal ☐ Property Damage

INCIDENT STATUS (HPG PROTOCOL)
Under Investigation

TRAFFIC CONDITION
☒ Heavy ☐ Normal ☐ Light

ROAD SURFACE CONDITION
☒ Asphalt ☐ Concrete ☐ Potholes

ROADWAY DIMENSIONS
Lane Width: Standard (3.0-3.5m)

TIRE EVIDENCE
Skid marks — 18m

VEHICLE MANEUVER
Loss of control

ROAD GEOMETRY
Straight / flat

CASUALTIES
2 injured, 0 fatal

OFFICER
Ramos, J. — PO2

BADGE
PO2-1089234

NOTES
SUV rolled after impact with truck. Debris scattered across lane 2. Skid marks indicate sudden braking.

EXPORT PDF

READ ONLY

Figure 15.4. Completed Incident Record View with Officer Parameters and Export Function

Completed Incident Record View with Officer Parameters and Export Function from figure 15.4 shows the same completed accident record, but the right panel is focused on the officer-verified information. It includes the scene ID and date, location, accident severity, incident status, traffic condition, road surface condition, roadway dimensions, tire evidence, vehicle maneuver, road geometry, casualties, officer name, badge number, and officer notes.

This screen is important because it combines **AI-generated evidence** with **officer-verified field information**. While ATADS assists in detecting and analyzing the scene, the officer still confirms and completes important investigation details. This supports a balanced system design where automation improves accuracy, but human judgment remains part of the documentation process.

The **Export PDF** button shows that the completed record can be converted into an official report. This supports the final documentation phase of ATADS, where the system produces standardized outputs for filing, review, and case reporting. The read-only status also helps ensure that the completed report remains consistent and protected after verification.

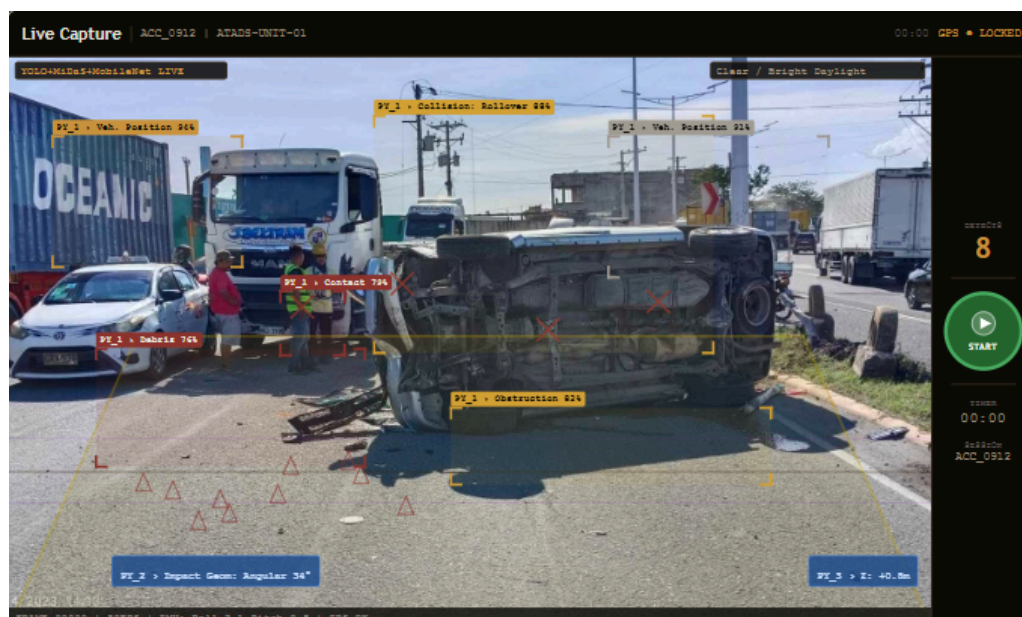


Figure 15.5. ATADS Live Capture Interface

Figure 15.5 shows the **Live Capture** screen used during active accident scene recording. The main area displays the real-time camera feed of the accident scene, with AI detection labels and bounding boxes shown directly on the image.

The system identifies important elements such as vehicle position, rollover, contact point, debris, obstruction, impact geometry, and elevation.

The right side of the screen shows the capture controls, including the **Start** button, GPS lock status, detection count, timer, and the current session ID. The top portion also shows the incident ID and ATADS unit number, which helps connect the live capture activity to the correct accident record and device. This interface is one of the most important parts of the prototype because it shows how ATADS replaces manual scene sketching with sensor-assisted digital documentation. Instead of relying only on the investigator's estimate, the system captures visual evidence, detects accident-related objects, and records sensor-based information. This helps reduce missing measurements, unclear visual references, and inconsistent accident scene interpretation.

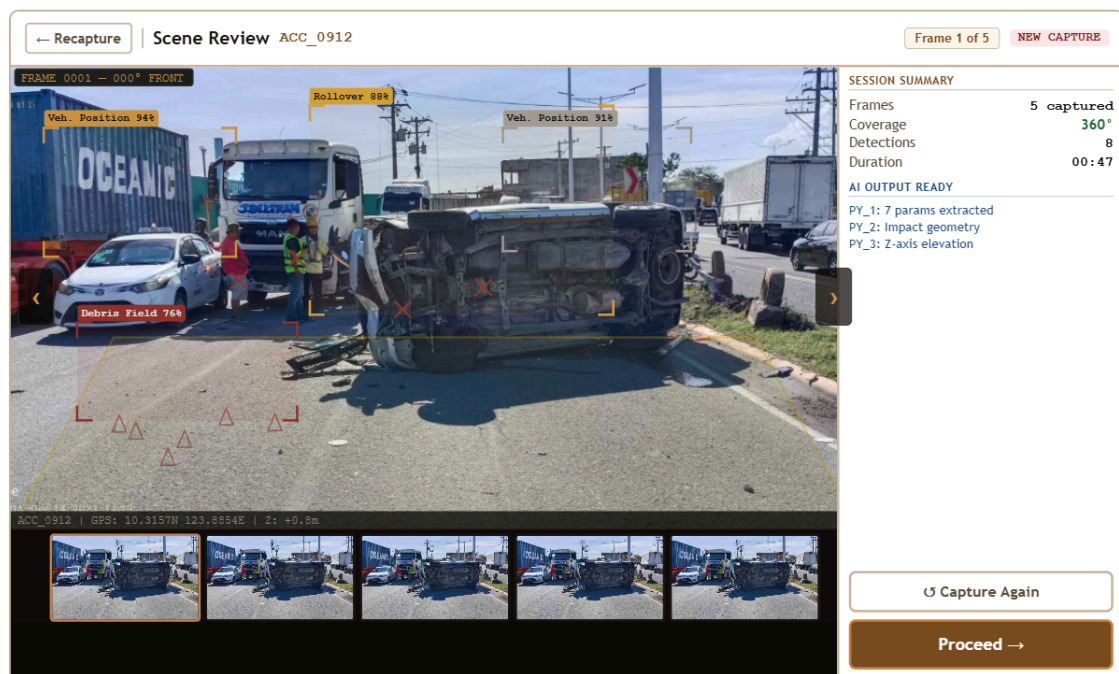


Figure 15.6. Scene Review Screen After Smart Capture

Figure 15.6 presents the **Scene Review** screen after the live capture session has been completed. It shows the captured accident scene with AI overlays and a strip of image frames at the bottom. The right panel provides a session summary, including the number of captured frames, coverage percentage, total detections, and capture duration.

The AI output section shows that the system has already extracted important analysis results from the captured scene. These include **PY_1 parameters**, **PY_2 impact geometry**, and **PY_3 z-axis elevation**. The screen also provides options to **Capture Again** or **Proceed**, giving the officer control over whether the captured data is acceptable or needs to be repeated.

This screen is important because it acts as a validation step before the incident documentation form is completed. It prevents incomplete or poor-quality data from moving forward in the workflow. If the coverage, detections, or image quality are not enough, the officer can capture the scene again. This supports the system's goal of improving documentation completeness and reliability.

Review

Incident Documentation ACC_0912

AI INPUTS - EDITABLE

PX-1 Camera Control Logic

VEHICLE POSITIONING

3 vehicles detected

DEBRIS FIELD

Lane 2 - est. 14m²

OBSTRUCTION MAPPING

Partial lane blockage

REFERENCE GRID

Active - 44x44 cells

COLLISION TYPE

Rollover

POINT OF CONTACT

Left-front quadrant

WEATHER / LIGHT

Clear / Bright Daylight

PX-2 GPS / NMEA

ACC_0912 | PX_1/2/3 auto-populated | PNP/TARAS

OFFICER INPUT - PNP/TARAS

SCENE ID

ACC_0912

DATE / TIME

Dec 14, 2023 14:38

GPS

10.3157N 123.8854E

HEADING

247° SW

LOCATION DESCRIPTION

Full location / landmark...

VEHICLE INVOLVEMENT

- Select -

ACCIDENT SEVERITY (ART. 365 RPC)

☐ Fatal
☒ Injury
☐ Property Damage

INCIDENT STATUS (HPG)

- Select -

TRAFFIC CONDITION

☐ Heavy
☒ Normal
☐ Light

ROAD SURFACE CONDITION

☒ Asphalt
☐ Concrete
☐ Potholes

ROADWAY DIMENSIONS

- Select -

OFFICER INPUT REQUIRED

Figure 15.7. Incident Documentation Form with AI Inputs and Officer Fields

Figure 15.7. Incident Documentation Form with AI Inputs and Officer Fields shows the **Incident Documentation** form for **ACC_0912**. The left panel displays AI-generated outputs, while the right panel contains fields that the officer must complete. These include scene ID, date and time, GPS coordinates, heading, location description, vehicle involvement, accident severity, incident status, traffic condition, road surface condition, and roadway dimensions.

The left panel shows that AI results are already available before the officer completes the form. These include vehicle positioning, debris field, obstruction mapping, reference grid, collision type, point of contact, and weather or lighting condition. This helps the officer by providing pre-analyzed information that can guide the completion of the official record.

This form is important because it connects automated system output with official investigation input. ATADS does not fully remove the role of the officer; instead, it assists the officer by reducing repetitive work and providing structured evidence. The form design supports standardized documentation by using text fields, dropdowns, and radio buttons instead of long free-text descriptions.

The screenshot displays the 'Incident Documentation' form for case ACC_0912. The interface is divided into two main columns. The left column contains several sections with pre-filled or auto-populated data: 'Partial lane blockage', 'REFERENCE GRID' (Active - 44x44 cells), 'COLLISION TYPE' (Rollover), 'POINT OF CONTACT' (Left-front quadrant), 'WEATHER / LIGHT' (Clear / Bright Daylight), 'GPS / NMEA' (PV-2), 'IMPACT GEOMETRY' (Angular - 34° approach), and 'IMU Interpreter' (Z-AXIS / ELEVATION: +0.8m offset from ref.). The right column contains dropdown menus for 'TRAFFIC CONDITION' (Normal selected), 'ROAD SURFACE CONDITION' (Asphalt selected), and 'ROADWAY DIMENSIONS'. Below these are input fields for 'TIRE EVIDENCE' (e.g. Skid 18m), 'CASUALTIES' (e.g. 2 inj, 0 fatal), 'OFFICER NAME' (Full name...), and 'BADGE / ID' (PO2-XXXXX). A large text area for 'OFFICER NOTES' contains the text 'Additional observations...'. At the bottom right is a prominent blue 'GENERATE REPORT' button. The footer shows 'ACC_0912 | PF_1/2/3 Auto-populated | PNP/TARAS' and a status indicator 'OFFICER INPUT REQUIRED'.

Figure 15.8. Incident Documentation Form with Report Generation Section

Figure 15.8 shows the lower portion of the incident documentation form. It includes additional required details such as tire evidence, casualties, officer name, badge number, and officer notes. At the bottom, the **Generate Report** button allows the system to create the final output, including the 2.5D diagram and PDF report.

This screen is important because it represents the final checking stage before report generation. The officer must enter or confirm important information that may

not be fully detected by the AI system, such as casualties, tire evidence details, and personal observations. These details are necessary because accident reports require both technical evidence and human observations.

The **Generate Report** function supports the main purpose of ATADS: to produce a standardized, organized, and evidence-based accident documentation output. Once the officer completes the form, the system can combine AI analysis, captured images, sensor readings, and officer notes into a formal report package. This improves the quality of traffic accident documentation and reduces the risk of missing important details.

The ATADS wireframes present a complete digital workflow for traffic accident documentation. The process begins with the startup screen, continues to the incident records dashboard, proceeds to live scene capture, moves through scene review, and ends with incident documentation and report generation. Each screen has a specific role in ensuring that accident data is properly captured, reviewed, completed, and stored.

The interface design supports the main objective of ATADS, which is to improve the accuracy, completeness, and standardization of traffic collision scene documentation. The system addresses common problems in manual documentation, such as missing measurements, unclear scene sketches, inconsistent symbols, and subjective interpretation. Through the use of camera capture, GPS, IMU data, AI detection, structured forms, and report generation, ATADS provides a clearer way to document traffic accident scenes.

Overall, these wireframes show that ATADS is designed not only as a recording tool but also as a guided investigation support system. It helps officers follow a clear process, reduces the chance of skipped information, and produces more consistent accident records. This makes the prototype relevant for traffic investigators, system developers, and researchers who aim to improve accident documentation through automation and digital evidence management

Hardware Requirements to Use

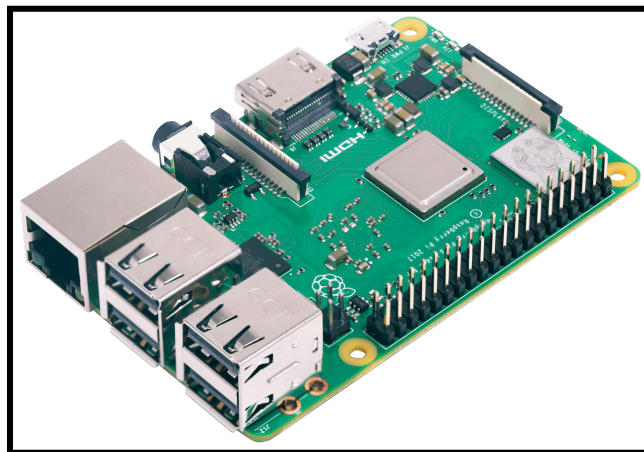


Figure 16. Raspberry Pi 4 Model B+ 2GB RAM

Figure 16 explains that in the context of the **ATADS** architecture, the **Raspberry Pi 4 Model B+** serves as the critical "Edge Computing" hub that bridges the physical accident scene with digital data. Its primary purpose is to act as a portable, low-power field controller that can simultaneously manage multiple hardware inputs—such as the NEO-6M GPS, the MPU6050 inertial sensor, and the camera module—without requiring a full-sized computer at the crash site. By using the Pi's GPIO (General Purpose Input/Output) pins, the system can capture real-time

telemetry at the exact moment of investigation, ensuring that data is recorded while the evidence is fresh and undisturbed.

The device is specifically used as a **Data Acquisition and Logging Unit**. While a standard microcontroller might struggle with handling high-resolution images alongside complex sensor streams, the Raspberry Pi 4 Model B+ has enough processing power and onboard memory to synchronize these disparate data sources. It handles the "First Responder" workload by running a lightweight interface on its attached monitor, allowing the user to verify that the sensors are calibrated and that the coordinates are accurate. This immediate verification is vital to solving the "Missing Measurements" and "Visual Clarity" issues identified in your research, as it prevents the responder from leaving the scene with incomplete data.

Ultimately, the Raspberry Pi 4 Model B+ is utilized to ensure **Data Integrity and Mobility**. Because it supports various communication protocols and has built-in storage capabilities, it can securely save the gathered evidence into a local directory before it is transferred to the PC for heavy analysis. Its compact form factor allows it to be mounted onto a handheld rig or a tripod, making it an ideal tool for navigating the often-congested and chaotic environment of a traffic collision. In short, the Pi provides the necessary "on-site intelligence" to transform a manual, error-prone sketching process into a standardized digital capture routine.

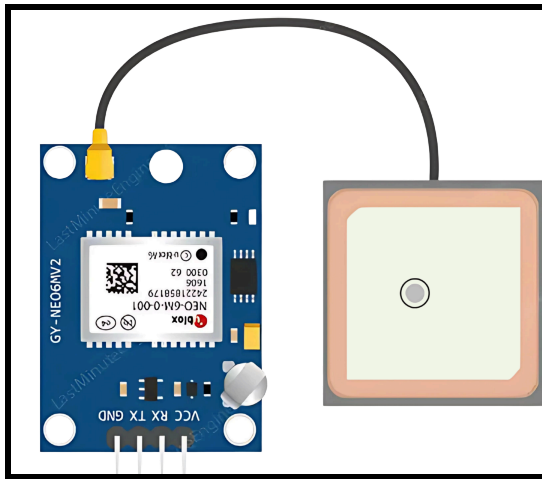


Figure 17. U-blox NEO-6m GPS w/ Active Antenna

Figure 17 shows the inclusion of the **U-blox NEO-6M GPS module equipped with an Active Antenna** is a critical technical requirement for the ATADS architecture, primarily serving as the system's primary source of **Geospatial Authentication**. In forensic engineering and accident reconstruction, the validity of a report is heavily dependent on the precision of its location metadata. The NEO-6M provides the necessary Global Navigation Satellite System (GNSS) telemetry to timestamp and geo-tag every captured data point, ensuring that the digital reconstruction is anchored to an immutable set of coordinates. This eliminates the "Standardization Gap" identified in research by replacing subjective, vernacular descriptions of location with standardized, globally recognized latitude and longitude values.

The "Active Antenna" component is particularly significant for the operational reliability of the system in varied environmental conditions. Unlike passive antennas, an active antenna incorporates an integrated **Low Noise Amplifier (LNA)** and a Surface Acoustic Wave (SAW) filter, which significantly enhances the signal-to-noise ratio (SNR) by amplifying weak satellite signals before they reach the receiver. This

is essential for traffic investigators who often operate in "Urban Canyons"—areas surrounded by tall buildings or dense foliage—where signal multipath interference can degrade GPS accuracy. The active antenna helps ensure a faster **Time-To-First-Fix (TTFF)** and maintains a stable lock on multiple satellites, which is vital for the high-precision requirements of a forensic diagramming system.

From a data-processing perspective, the NEO-6M utilizes the **NMEA-0183 protocol** to communicate with the Raspberry Pi 4 Model B+ via UART. This allows the system to parse and store "Sentence" data (such as GPRMC and GPGLA) which contain not only coordinates but also altitude, satellite count, and atomic-clock synchronized timestamps. Within the ATADS framework, this data acts as a digital seal of integrity. By hard-coding these technical parameters into the final report, the system directly overcomes the "Phantom Evidence" challenge documented in your study, as the location data is generated by a verified hardware source rather than being reconstructed from the investigator's potentially flawed memory or manual field notes.

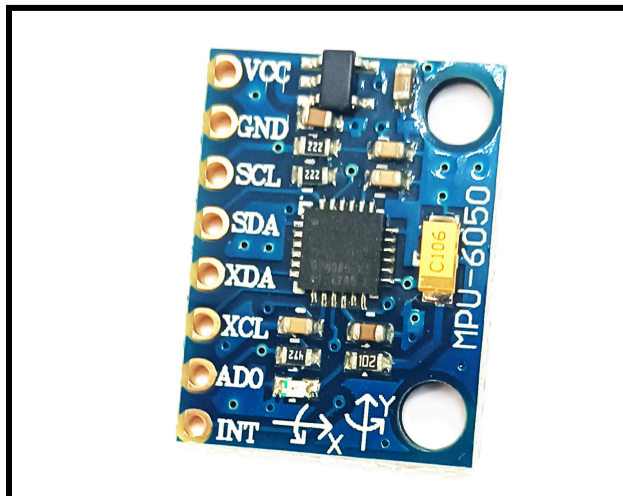


Figure 18. MPU6050 (Gyroscope & Accelerometer)

The **MPU6050** in figure 18, which integrates a **3-axis Gyroscope** and a **3-axis Accelerometer**, is a fundamental component of the ATADS architecture, providing the system with **Spatial Orientation and Inertial Telemetry**. In the context of accident reconstruction, capturing the visual scene is insufficient if the orientation of the recording device is unknown. The MPU6050 allows the system to establish a local coordinate frame, measuring the "Pitch," "Roll," and "Yaw" of the hardware unit during the data acquisition phase. This precise orientation data is crucial for correcting image distortion and ensuring that 3D renders are accurately aligned with the gravitational horizon, thereby eliminating the "Visual Clarity" issues identified in your research.

The accelerometer component of the MPU6050 serves as a **Tilt Sensing mechanism**, detecting the device's alignment relative to the Earth's gravity. When a responder is capturing data on uneven terrain—a common occurrence at roadside accident scenes—the MPU6050 records the exact angle of the slope. This data is fed into the **PC/Laptop processing station**, where algorithms use the inertial data to "level" the digital reconstruction. By mathematically compensating for the angle at which the data was gathered, the system prevents the spatial distortions and "Missing Measurement" errors that often occur when investigators attempt to manually estimate distances on inclined surfaces.

Furthermore, the MPU6050 utilizes an on-chip **Digital Motion Processor (DMP)** to handle complex calculations, such as fusing raw sensor data into quaternions or Euler angles via I2C communication. This hardware-level processing reduces the computational burden on the Raspberry Pi 4 Model B+, allowing for real-time monitoring of system stability. Within the ATADS framework, this sensor serves as a safeguard against human error; if the responder's movements are too

erratic or if the device is tilted beyond a certain threshold, the system can provide immediate feedback. This helps ensure that the evidence collected is geometrically sound, directly addressing the "Standardization Gap" by providing a level of objective, mathematical precision that manual investigation cannot achieve.

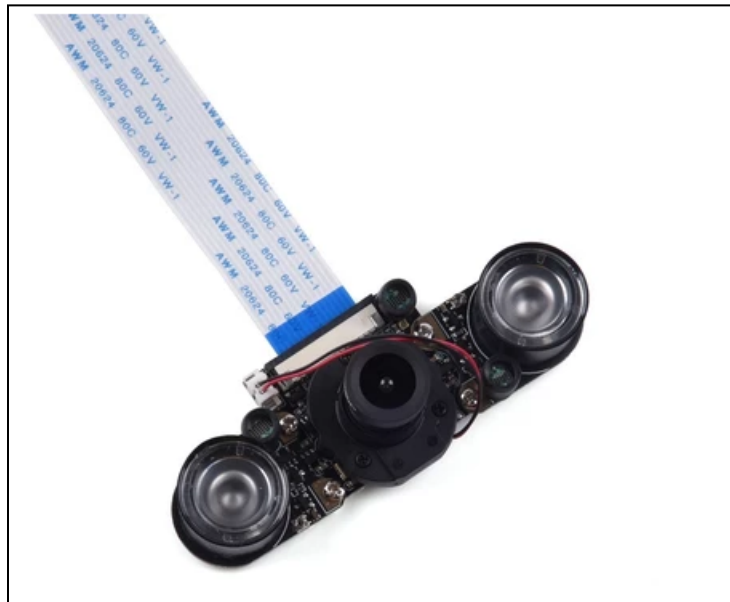


Figure 19. Raspberry Pi Day/Night Vision Camera Module

Figure 19 shows the **Raspberry Pi Day/Night Vision Camera Module with a 75-degree field of view** as the main optical data acquisition component of the Automated Traffic Accident Documentation System (ATADS). In accident scene documentation, the field of view is important because it determines how much of the scene the camera can capture in one frame. A camera with a smaller viewing angle may require the investigator to stand farther away to capture the full collision area. However, this is not always possible because of traffic, road debris, damaged vehicles, pedestrians, or other road obstructions. The 75-degree field of view provides a wider viewing area than a standard narrow-angle camera while still

maintaining a focused image. This allows the system to capture vehicle positions, road markings, skid marks, and surrounding reference points from a practical distance. This is relevant to the study because visual clarity and blurred images were identified as one of the investigation barriers. By capturing a clearer and wider view of the accident scene, the camera helps reduce the risk of missing important visual evidence.

From a technical perspective, the camera provides the raw image data needed by the PC or laptop for digital accident documentation. When the ATADS software creates a digital diagram or supports scene reconstruction, it needs clear images with enough reference points. These reference points may include lane markings, road edges, skid marks, vehicle corners, traffic signs, and other fixed objects near the accident area. The 75-degree field of view helps include these details in a single frame, making it easier for the system to compare and align the visual data with the information gathered from other sensors. By capturing more scene details with fewer images, the system can also reduce the possibility of missing measurements, which is a common problem in manual accident investigation.

Furthermore, the camera module can be integrated with the Raspberry Pi through the camera interface, allowing the system to capture image data efficiently during field operation. This is important because traffic accident investigators often work under time pressure, especially when accidents occur on active roads. The camera helps shorten the time needed to document the scene because it can cover a useful viewing area in each capture. Within the ATADS framework, the camera serves as the visual foundation of the system. The images captured by the camera

can be combined with GPS data and MPU6050 sensor data to create a more complete and standardized accident record. This supports the main purpose of ATADS, which is to reduce the limitations of hand-drawn sketches, improve documentation accuracy, and provide a clearer digital record of the accident scene.

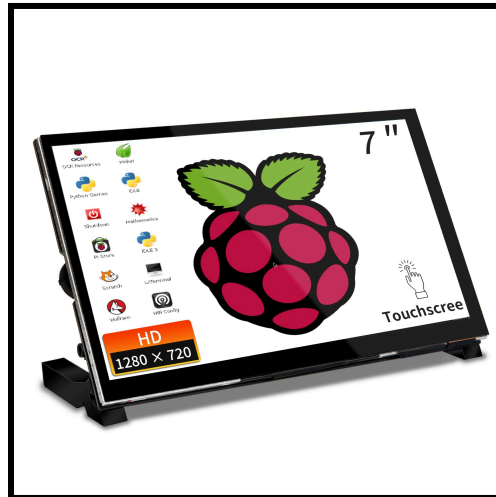


Figure 20. Raspberry Pi 5-inch Touchscreen Monitor

Figure 20 explains that the **Raspberry Pi 5-inch Touchscreen Monitor** serves as the primary **Human-Machine Interface (HMI)** for the field acquisition unit, providing the necessary visual feedback and control for on-site responders. In a manual investigation environment, responders often lack immediate verification of the data they are recording, leading to the "Missing Measurements" and "Phantom Evidence" identified in your research. By integrating a dedicated touchscreen, the ATADS prototype grants the user a real-time "Viewfinder" and system dashboard, allowing them to confirm that the camera is focused, the GPS has a valid lock, and the MPU6050 is properly calibrated before the investigator leaves the scene.

Technically, the monitor utilizes the **DSI (Display Serial Interface)** port on the Raspberry Pi 4 Model B+, which allows for high-speed data transfer without occupying the HDMI port or significant CPU overhead. The capacitive 10-finger touch capability is essential for the "Utility-First" design philosophy of the ATADS software; it enables a button-free interface where responders can navigate through setup wizards, input incident metadata, and trigger data captures via large, touch-friendly icons. This is particularly vital under the "**High Time Pressure**" documented in your study, as a touchscreen interface is significantly faster and more intuitive to operate in a chaotic roadside environment than a traditional keyboard and mouse setup.

Beyond navigation, the monitor acts as a **Field Validation Tool**. After a capture is completed, the screen allows the responder to perform an immediate "Quality Assurance" check of the gathered images and sensor logs. This capability directly addresses the finding that **67.6% of interpretation barriers** stem from poor visual clarity; if a captured image is blurred or a sensor reading appears anomalous on the 5-inch display, the responder can immediately re-capture the data. This "closed-loop" feedback system helps ensure that only high-quality, standardized evidence is transferred to the **PC/Laptop Analysis Station**, thereby guaranteeing the integrity of the final accident diagram.

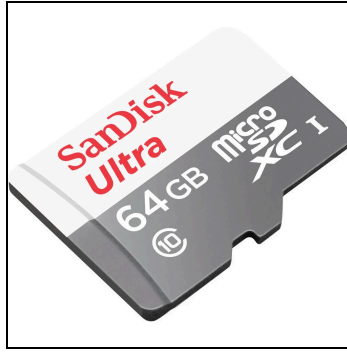


Figure 21. 64gb MicroSD Card

Figure 21 in the ATADS architecture, the **64GB MicroSD Card** serves as the **Primary Non-Volatile Storage Medium** for the Raspberry Pi 4 Model B+. While the sensors and camera gather raw data, the MicroSD card is the physical repository where this information is committed to memory. In the context of your research findings, where "Missing Measurements" and inconsistent data were primary barriers, the MicroSD card helps ensure that once a data point is captured, it is permanently logged and protected from volatile memory loss, providing a reliable "Chain of Custody" for digital evidence.

From a technical perspective, the capacity and speed of the card are vital for handling the **High-Bandwidth Data Streams** generated during an investigation. A 75-degree camera module produces high-resolution image files, which, when combined with continuous telemetry logs from the MPU6050 and NEO-6M GPS, can consume significant storage space over multiple incidents. A 64GB capacity (specifically an **SDHC Class 10** or higher) is required to ensure sufficient "Write Speeds," allowing the Raspberry Pi to save data rapidly without causing system lag or buffer overflows. This speed is essential to meet the "High Time Pressure" demands identified in your study, as it minimizes the "Save Time" between captures.

Furthermore, the MicroSD card acts as the **Physical Data Bridge** in your block diagram's "Data Transfer" phase. Since the Raspberry Pi 4 model B+ is a field unit and the heavy processing occurs on a PC/Laptop, the MicroSD card (or a secondary USB drive initialized via the card) allows for the secure, offline migration of evidence. This "Air-Gapped" transfer method helps ensure that even in areas with zero internet connectivity—a common scenario at remote accident sites—the responder can physically transport the standardized data to the investigator. This hardware choice guarantees that the "Visual Clarity" and precise measurements captured in the field remain uncorrupted during the transition to the final processing station.

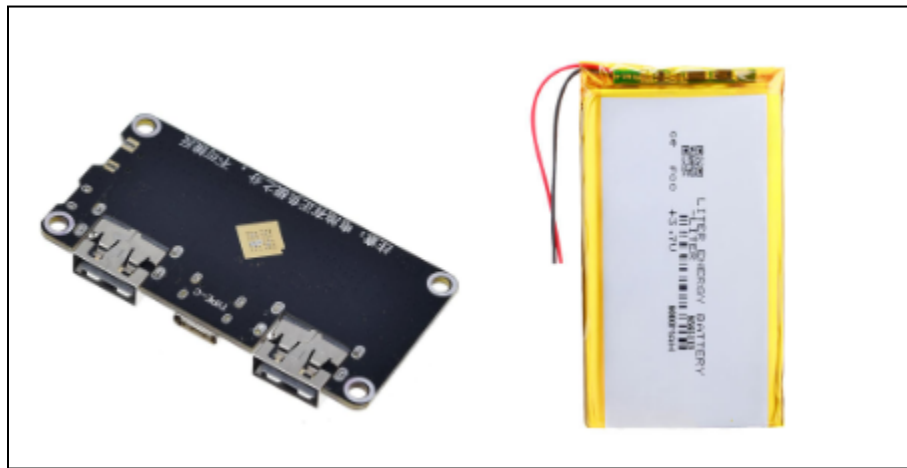


Figure 22. IP5328P Dual USB Module and 8565113 LiPo Battery

Figure 22 explains that the IP5328P Dual USB Module serves as the primary power management and voltage regulation unit of the ATADS field system. In line with the research findings that emphasize “High Time Pressure” and unpredictable roadside environments, this module helps ensure that the system delivers a stable

5V output with sufficient current to meet the Peak Power Demand of the Raspberry Pi 4 Model B+ and its peripherals. The simultaneous operation of the touchscreen display, camera module, and NEO-6M GPS introduces a significant load, requiring consistent voltage regulation to avoid under-voltage conditions or “Brownouts.”

Through its integrated boost converter and power management circuitry, the IP5328P maintains stable system operation, preventing corrupted sensor logs, incomplete measurements, or sudden shutdowns. This stability directly supports the ATADS objective of resolving “Visual Clarity” and “Missing Measurements” issues by ensuring uninterrupted sensor performance throughout the data acquisition process.

The 8565113 LiPo Battery functions as the primary energy source of the ATADS field unit, enabling fully autonomous and portable system operation. Given that accident scenes often lack access to fixed power infrastructure, the battery helps ensure that the Raspberry Pi 4 Model B+ and all connected sensors remain operational during the entire investigation process, eliminating “Data Gaps” caused by power interruption.

In addition to sustaining system uptime, the LiPo battery enables the Operational Mobility required for effective field deployment. By removing dependence on stationary or vehicle-based power sources, the responder can freely move around the accident scene and capture data from optimal angles. This mobility is critical in reducing “subjective interpretation” in manual sketches, as it allows precise positioning of sensors such as the camera and MPU6050 to capture accurate and objective evidence. Within the ATADS framework, the battery acts as

the core energy enabler that supports a mobile and reliable forensic documentation system.



Figure 23 Ulanzi VL49 RGB LED Video Light

Figure 23 shows the inclusion of the Ulanzi VL49 RGB LED Video Light as a critical technical requirement for the ATADS architecture, primarily serving as the system's controlled illumination source for image acquisition. In computer vision-driven accident reconstruction, the reliability of detection, segmentation, and spatial interpretation is fundamentally dependent on the consistency of visual input data. The illumination module provides a uniform and stabilized lighting environment that enables accurate object segmentation, depth estimation, and feature extraction, ensuring that each captured frame maintains consistent exposure characteristics. This directly addresses the "Visual Ambiguity" and "Structural Flaws" identified in the study by minimizing shadow artifacts, glare distortions, and low-light noise, thereby replacing inconsistent environmental illumination with a standardized and controlled lighting baseline.

From a functional standpoint, the illumination module operates as a supporting subsystem within the ATADS field-tier architecture, augmenting the performance of the Raspberry Pi Day/Night Vision Camera Module during the data acquisition phase. The device delivers adjustable brightness output and a variable color temperature range between 2500K and 9000K, allowing the system to adapt to diverse environmental conditions such as nighttime scenes, low-visibility areas, and uneven ambient lighting. Its high Color Rendering Index ($CRI \geq 95$) helps ensure accurate color fidelity of scene elements, including debris distribution, skid marks, and vehicle surface conditions, which are essential for evidentiary analysis and reconstruction accuracy. By preserving the true visual characteristics of captured data, the module enhances the reliability of downstream inference processes executed by YOLOv8n-seg and MiDaS Small within the ATADS pipeline.

Structurally, the illumination unit is positioned in close proximity to the camera subsystem to achieve optimal light distribution across the field of view while minimizing occlusion and directional shadowing. This configuration is critical during the smart capture session, where multiple frames are acquired from varying angles, as it helps ensure that illumination conditions remain consistent across successive captures. The module operates using an internal rechargeable power source and may be supplemented through USB connectivity, allowing flexible integration without imposing significant load on the primary system power supply. Through the incorporation of a dedicated illumination subsystem, ATADS mitigates environmental lighting variability, enhances data acquisition fidelity, and reinforces the system's objective of producing standardized, high-quality, and forensically reliable accident scene documentation.

Software Technologies to Use



Figure 24. Python 3.12

Figure 24 explains that Python 3.12 is selected as the primary programming language for the field acquisition unit due to its high-level abstraction and extensive library support for hardware communication. Python's ecosystem provides robust, well-documented modules such as Pynmea2 and SMBus2 which allow for seamless I2C and UART interfacing. This efficiency helps ensure that the Raspberry Pi 4 Model B+ can act as a high-speed data logger, capturing and timestamping evidence without the overhead associated with lower-level languages, which is critical during the high-pressure environment of a traffic investigation.

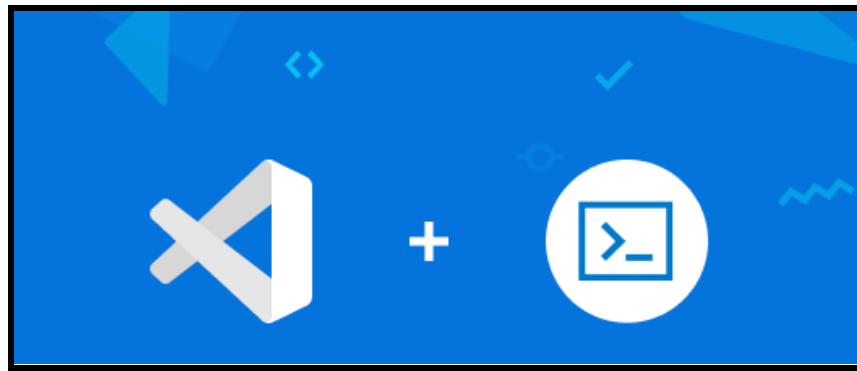


Figure 25. VS Code (Remote-SSH)

Figure 25 shows that the project utilizes **VS Code with the Remote-SSH extension** as the primary development environment rather than the built-in Thonny IDE. While Thonny is sufficient for basic scripts, the ATADS project involves a complex architecture that benefits from the professional-grade features of VS Code, such as advanced IntelliSense, integrated Git version control, and multi-file management. By using Remote-SSH, developers can author and debug code on a high-performance laptop while the scripts execute directly on the Raspberry Pi's hardware. This workflow is significantly more efficient than coding directly on the Pi, as it eliminates the lag of the Pi's graphical desktop and allows for a more comfortable, full-sized development workspace while maintaining real-time hardware access.

Work Breakdown Structure

A. Hardware Design and Fabrication

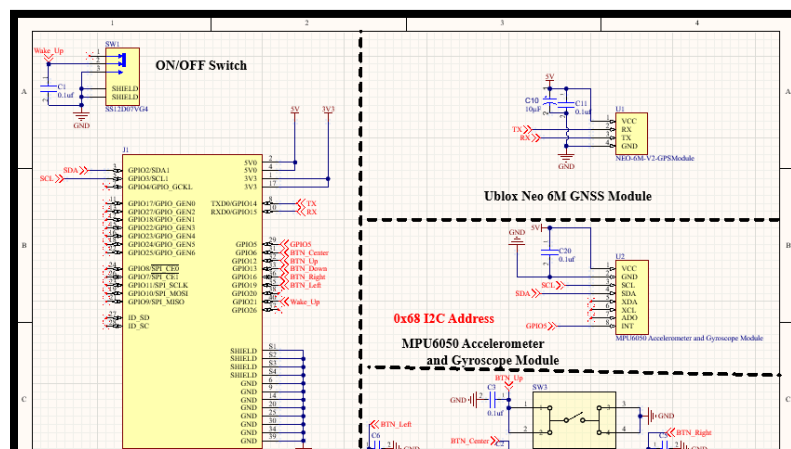


Figure 26. ATADS Schematic Diagram

The schematic diagram of the Autonomous Traffic Accident Detection System (ATADS), as illustrated in Figure 26, presents the complete electrical interconnection of all hardware components that constitute the system's core architecture. At the center of the design is the Raspberry Pi 4 Model B+, which serves as the primary processing unit responsible for coordinating data acquisition, processing, and communication across all peripheral modules. The microprocessor interfaces with a variety of input and output components through its General-Purpose Input/Output (GPIO) pins, enabling flexible and programmable hardware control.

The system integrates a Ublox Neo 6M GNSS Module, positioned in the upper-right section of the schematic, which provides real-time geographic positioning data essential for accurately determining the location of a detected traffic incident. Communication between the GNSS module and the Raspberry Pi is established through a serial UART interface, ensuring reliable and continuous data transmission. In parallel, an MPU-6050 Accelerometer and Gyroscope Module is incorporated into

the design, operating at the designated I2C address 0x68. This sensor is responsible for capturing six-degrees-of-freedom motion data, including linear acceleration along three axes and angular velocity, which are critical parameters for detecting sudden vehicular impacts or abnormal motion events characteristic of road accidents.

Power management within the system is handled through a dedicated ON/OFF switch circuit, accompanied by pull-up resistors and bypass capacitors that ensure stable voltage regulation across all components. The lower section of the schematic further illustrates an array of control buttons interfaced through GPIO lines, which facilitate manual interaction and operational mode selection. Additionally, RTC (Real-Time Clock) circuitry and I2C pull-up resistor networks are incorporated to maintain accurate timekeeping and ensure proper signal integrity across the I2C communication bus. The comprehensive nature of this schematic reflects a well-structured hardware design philosophy that prioritizes modularity, signal integrity, and system reliability.

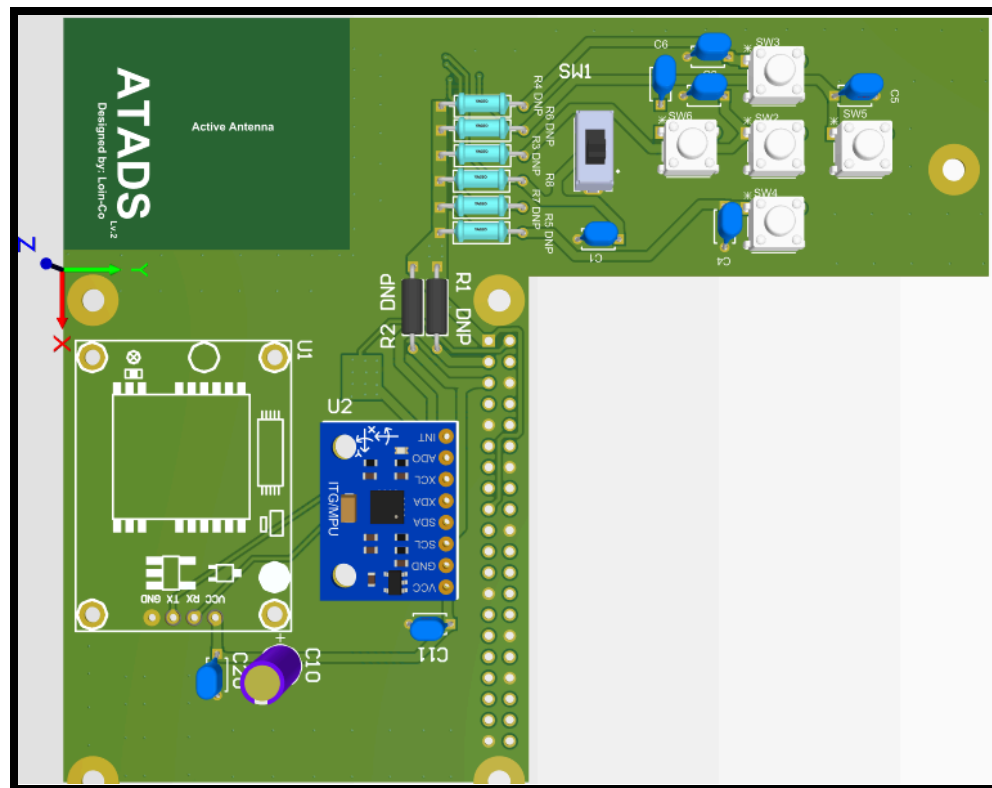


Figure 27. PCB Design for ATADS with components

Figure 27 presents the physical realization of the ATADS hardware through its Printed Circuit Board (PCB) layout, rendered with all electronic components populated in their designated positions. The PCB design follows a compact, L-shaped form factor, which optimizes the spatial arrangement of components while maintaining accessibility for connectors and user interface elements. The board is designed on a standard two-layer green substrate, with component placement strategically distributed to minimize signal interference and reduce trace lengths for high-frequency communication lines.

The upper-right region of the PCB accommodates the control button array (SW1) along with supporting passive components such as capacitors and resistors, providing the user with a direct hardware interface for operational control. The active antenna designated for the GNSS module is visibly allocated in the upper-left area, ensuring unobstructed signal reception critical for accurate geographic positioning.

The Raspberry Pi interface module (U1) is mounted on the left side of the board, serving as the central hub to which all peripheral components are electrically connected. Adjacent to it, the MPU-6050 inertial measurement unit (U2) is mounted and clearly identifiable by its distinctive blue module, positioned to facilitate consistent motion sensing with minimal mechanical obstruction.

Passive components, including decoupling capacitors (C10, C11) and resistor networks (R1, R2 marked as DNP — Do Not Populate), are distributed across the board in accordance with best practices for electromagnetic compatibility and noise suppression. The DNP designations indicate provisions for optional components that may be selectively installed depending on specific deployment requirements or design revisions. Overall, the PCB layout demonstrates a methodical and professional approach to embedded system hardware integration, balancing functional density with design maintainability and ease of assembly.

B. Embedded Software and Acquisition Logic

The embedded software layer of the ATADS is designed to operate in direct correspondence with the hardware architecture established in the schematic and PCB design phases. Running on the Raspberry Pi 4, the acquisition logic is responsible for initializing and continuously polling the MPU-6050 sensor via the I2C protocol, retrieving raw accelerometer and gyroscope data at a defined sampling

rate to ensure timely detection of anomalous motion events. Concurrently, the software manages serial communication with the Ublox Neo 6M GNSS module, parsing NMEA sentence strings to extract valid latitude, longitude, and timestamp information that are appended to any triggered event log.

The acquisition logic employs threshold-based detection algorithms, wherein acceleration magnitudes exceeding predefined values — indicative of collision-level forces — trigger an event flag within the software pipeline. Upon detection, the system consolidates the inertial data with the corresponding GPS coordinates and RTC timestamp, forming a structured data packet that can be transmitted, stored, or further processed. The software architecture is organized in a modular fashion, separating sensor drivers, data processing routines, and communication handlers into discrete functional layers, thereby facilitating maintainability, scalability, and systematic debugging throughout the development lifecycle.

Risk Factors and Preventive Measures

The Risk Factors and Preventive Measures identify possible problems that may affect the accuracy, completeness, and reliability of ATADS during prototype development and field use. Since the system depends on camera capture, GPS, IMU, AI-assisted outputs, officer input, database storage, and report generation, each risk must be controlled to protect the quality of traffic accident documentation.

Table 8. Scene Capture and Evidence Collection Risks

Risk Factor	Preventive Measure
Incomplete accident scene capture	Use scene review, frame summary, and capture-again option.

Missing vehicle position data	Require multi-angle capture and AI-assisted vehicle positioning.
Missed debris or skid evidence	Use AI detection overlay and officer verification.
Blurry or low-quality image capture	Review captured frames before proceeding to documentation.
Poor lighting or weather condition	Record weather/light conditions and allow manual validation.
Scene obstruction by people or vehicles	Require capture from multiple angles and repeat unclear captures.

This category focuses on the risk of collecting weak or incomplete accident evidence during field documentation. Since ATADS depends on camera-based capture and multi-angle scene recording, incomplete frames, blurred images, blocked views, poor lighting, or missed debris and skid marks can reduce the quality of the final report. The preventive measures strengthen this stage by requiring scene review, capture-again options, AI detection overlays, and officer verification before the documentation proceeds. This helps ensure that the accident scene is not accepted too early and that important evidence is checked while the officer is still on-site.

Table 9. GPS, IMU, and Sensor Accuracy Risks

Risk Factor	Preventive Measure
Wrong GPS location reading	Check GPS lock before capture and require manual location confirmation when needed.
Weak or unstable GPS signal	Display GPS status and mark uncertain location data for review.
Incorrect IMU orientation reading	Require IMU readiness checking and calibration before capture.

Wrong elevation or road
slope interpretation

Use IMU-supported z-axis data with officer
confirmation.

This category covers the risks related to location, orientation, and elevation accuracy. ATADS uses GPS to support location reference and IMU data to support device orientation, road tilt, and z-axis interpretation, so inaccurate sensor readings may affect the generated diagram and scene reconstruction. The preventive measures require GPS lock checking, manual confirmation for uncertain locations, IMU readiness checking, and officer review of elevation-related outputs. These controls are important because the system should not rely on unverified geospatial or inertial data when producing an accident diagram.

Table 10. AI Detection and Analysis Risks

Risk Factor	Preventive Measure
AI misclassification of accident elements	Treat AI output as support data and require officer verification.
Low-confidence AI detection	Review uncertain AI labels before final report generation.
Overdependence on AI results	Keep officer judgment required in the documentation form.

This category addresses the risk that AI-assisted outputs may be incomplete, uncertain, or incorrectly classified. ATADS uses AI to support detection of accident-related elements such as vehicles, debris, obstruction, collision type, contact point, impact geometry, and elevation-related information, but these outputs should not be treated as final evidence without human review. The preventive measures ensure that AI is used as a decision-support tool instead of a replacement

for the officer. By requiring officer verification, ATADS maintains professional judgment and reduces the risk of accepting incorrect automated results.

Table 11. Officer Input and Documentation Risks

Risk Factor	Preventive Measure
Missing officer input	Use required fields before report generation.
Wrong manual data entry	Use dropdowns, radio buttons, and standardized selections.
Skipped accident severity or casualty details	Require completion of severity and casualty fields.

This category focuses on the human-entry part of the documentation process. Even if ATADS captures the scene correctly, the final report may still become incomplete if the officer skips important fields or enters inconsistent details. Required fields, dropdowns, radio buttons, and standardized selections help reduce missing or unclear information, especially for accident severity, casualty details, road condition, vehicle involvement, and officer notes. These measures support clearer reporting because the system guides the officer to complete essential details before generating the final output.

Table 12. Record Linking and Evidence Integrity Risks

Risk Factor	Preventive Measure
Incorrect record linking	Use unique incident ID, session ID, timestamp, and officer ID.
Accidental editing of completed records	Apply read-only mode for finalized cases.
Loss of evidence integrity	Connect every report to officer details, capture sessions, AI outputs, and logs.

This category deals with the integrity and traceability of accident records. A serious risk in a documentation system is when captured images, AI outputs, officer inputs, and final reports are linked to the wrong incident record or changed after completion. Using unique incident IDs, session IDs, timestamps, officer IDs, and read-only protection helps maintain a clear evidence trail from scene capture to final report. This is important because traffic accident records must remain reviewable, organized, and protected from accidental changes after they are finalized.

Table 13. Report Generation and Output Risks

Risk Factor	Preventive Measure
Report generation failure	Validate incident details, AI analysis, captured frames, and officer inputs before generating.
Incomplete PDF report output	Review completed record before export.

This category focuses on the final output stage of ATADS. The report is the main deliverable of the system, so failure to generate the report or export a complete PDF can weaken the whole documentation process. Validating incident details, captured frames, AI analysis, and officer inputs before generation helps ensure that the report contains the needed evidence and verified information. Reviewing the completed record before export also allows the officer to confirm that the final document is complete, readable, and ready for filing or review.

Table 14. Storage and System Log Risks

Risk Factor	Preventive Measure
Lost captured frames or report files	Check storage availability and save session data immediately.

SD card or storage failure	Maintain backups and system logs for saved outputs.
Missing system activity trail	Record startup, GPS lock, capture start/stop, report generation, and errors.

This category addresses the risk of losing captured evidence, report files, or system activity records. Since ATADS saves scene frames, AI detections, report outputs, and logs, storage failure or missing logs may make it difficult to verify what happened during a capture session. Checking storage availability, saving data immediately, maintaining backups, and recording key system actions help protect the documentation trail. These measures also support troubleshooting because system logs can show whether GPS locked, capture started, capture stopped, reports were generated, or errors occurred.

Table 15. Performance and Power Risks

Risk Factor	Preventive Measure
Slow live capture or AI processing	Use lightweight models and preload AI modules during startup.
Power interruption during field use	Check battery or power source before operation.

This category focuses on the operational reliability of the prototype during field use. ATADS processes camera frames, sensor data, AI detection, scene review, and report generation, so slow performance may delay the officer during time-sensitive accident documentation. Power interruption is also a major risk because the device may shut down before the scene is fully captured or before the report is saved. Using lightweight AI models, preloading modules, and checking the

battery or power source before operation helps keep the system responsive and stable during documentation.

Table 16. User Interface and Field Usability Risks

Risk Factor	Preventive Measure
Touchscreen misclicks during field use	Use large buttons, clear labels, and confirmation prompts.
Confusing interface workflow	Maintain a step-by-step flow from dashboard to report generation.

This category covers risks related to user interaction, especially because ATADS may be used in stressful roadside conditions. If buttons are small, labels are unclear, or the workflow is confusing, the officer may press the wrong option, skip a step, or spend more time completing the documentation. Large buttons, clear labels, confirmation prompts, and a step-by-step workflow help make the system easier to operate. These measures support faster and more confident use of the prototype from dashboard, capture, review, documentation, report generation, and export.

Table 17. Security and Prototype Acceptance Risks

Risk Factor	Preventive Measure
Unauthorized access to records	Limit system use to authorized officers and badge-linked accounts.
Prototype rejection due to low practicality	Conduct usability testing and revise the interface based on user feedback.

This category focuses on system protection and user acceptance. Accident records may contain sensitive case details, officer information, location data, captured images, and generated reports, so access should be limited to authorized users. At the same time, the prototype may be rejected if it is technically functional but difficult to use in actual field conditions. Badge-linked access, usability testing, and interface revision based on user feedback help protect the system while also improving practicality, trust, and acceptance among intended users.

System Test Plan

The System Test Plan serves as the quality checkpoint of the Automated Traffic Accident Documentation System (ATADS). It verifies whether the prototype can perform its intended function from the moment it opens, records an incident, captures the accident scene, processes AI-assisted outputs, accepts officer inputs, and generates the final report. Since ATADS is designed to reduce missing measurements, unclear visual references, scale errors, and inconsistent manual documentation, the test plan focuses on proving that each major screen and function works as part of one complete accident documentation workflow.

ATADS is not tested only as a software interface. It is tested as a field-support documentation tool where the camera, GPS, IMU, AI modules, record dashboard, review screen, officer form, and report output must work together. The goal of the test plan is to determine whether the prototype can guide the investigator from raw scene capture to a structured and standardized traffic collision record without losing important data.

Table 8. System Test Plan for ATADS Prototype

Test No.	Test Area	Test Procedure	Expected Result
1	Welcome and Startup Screen	Open the ATADS prototype and observe the startup interface. Tap the screen to proceed.	The system displays the ATADS welcome screen, shows the system status, and allows the user to continue to the main dashboard.
2	Incident Records Dashboard	Open the main incident records screen. Check the record counters, table layout, and status labels.	The system displays the total records, records in review, drafts, completed records, and the incident table with scene ID, date and time, location, collision type, vehicle involvement, and status.
3	Record Search Function	Enter an incident ID or keyword in the search bar.	The system filters the records and displays the matching incident information accurately.
4	Unit Status Display	Check the unit status panel for GPS, IMU, camera, and AI models.	The system clearly shows whether GPS is locked, IMU is ready, the camera is active or on standby, and the AI models are loaded.
5	New Capture Access	Select the Capture button from the dashboard.	The system opens the Live Capture screen and prepares a new recording session under the correct incident and ATADS unit.
6	Live Camera Feed	Start the capture session and observe the accident scene display.	The system shows a clear live camera view of the accident scene and displays the current session details such as incident ID, unit ID, timer, and GPS status.
7	AI Detection Overlay	Allow the system to analyze the live scene during capture.	The system displays AI-assisted labels and visual markers for detected accident elements such as vehicle position, collision type, contact point, debris field, obstruction, impact geometry, and elevation.

8	GPS Reading	Capture a scene while GPS is locked.	The system records the GPS coordinates and connects them to the active accident record.
9	IMU Reading	Capture a scene while the device is positioned during field use.	The system reads orientation data such as roll, pitch, heading, or elevation reference, helping reduce distortion in the accident scene output.
10	Capture Completion	Complete the live capture session.	The system proceeds to the Scene Review screen and displays the captured frame set.
11	Scene Review Summary	Review the session summary after capture.	The system shows the number of captured frames, coverage percentage, detection count, and duration, allowing the officer to decide whether the capture is acceptable.
12	Capture Again Function	Select Capture Again from the Scene Review screen.	The system allows the officer to repeat the capture when the result is incomplete, unclear, or insufficient.
13	Proceed to Documentation	Select Proceed after accepting the scene review.	The system opens the Incident Documentation form and carries forward the AI-assisted scene outputs.
14	AI Output Panel	Check the AI-generated output panel in the documentation screen.	The system displays structured outputs from PY_1, PY_2, and PY_3, including vehicle positioning, debris field, obstruction mapping, collision type, point of contact, impact geometry, and z-axis/elevation data.
15	Officer Input Fields	Fill in the required officer and incident details.	The system accepts standardized inputs such as location, vehicle involvement, accident severity, incident status, traffic condition, road surface, roadway dimensions, tire evidence, casualties, officer name, badge number, and notes.

16	Standardized Selection Fields	Use dropdowns and radio buttons in the form.	The system allows the officer to select consistent categories instead of relying only on long manual descriptions.
17	Required Field Checking	Attempt to generate a report while required fields are incomplete.	The system prevents incomplete report generation and guides the officer to complete the missing information.
18	Generate Report	Complete the documentation form and select Generate Report .	The system generates the final report output containing the accident details, AI-assisted findings, officer inputs, and diagram/report package.
19	Completed Record View	Open a completed incident record from the dashboard.	The system displays the completed record with captured images, AI analysis, officer parameters, and record status.
20	Read-Only Protection	Open a completed record marked as read-only.	The system allows viewing and exporting but protects finalized details from accidental changes.
21	Export PDF	Select Export PDF from the completed record view.	The system exports the completed accident documentation into a PDF report for filing and review.
22	Data Storage	Save a completed incident record after report generation.	The system stores the incident details, capture session, scene frames, AI detections, AI analysis, officer information, and generated report in the database structure.
23	System Logs	Perform key actions such as startup, GPS lock, capture start, capture stop, record creation, and report generation.	The system records important activities in the system logs for monitoring and traceability.

24	End-to-End Workflow	Perform the complete process from startup to dashboard, live capture, scene review, incident documentation, report generation, and completed record viewing.	The system completes the full ATADS workflow without data loss, incorrect record linking, or interruption.
----	---------------------	--	--

The System Test Plan evaluates whether ATADS can complete the full traffic accident documentation workflow in a clear, reliable, and organized manner. The test begins with the welcome screen and incident records dashboard to confirm that the system can properly open, display case summaries, show record statuses, and provide access to existing and new accident records. It then checks the unit status panel to ensure that the GPS, IMU, camera, and AI models are properly monitored before field capture begins. This is important because ATADS depends on these components to collect accurate location data, orientation readings, visual evidence, and AI-assisted scene outputs. The live capture tests further verify if the system can display the accident scene, identify key elements such as vehicle position, collision type, debris, obstruction, contact point, impact geometry, and elevation, and connect these findings to the correct incident record.

The test plan also checks the review, documentation, and reporting stages to ensure that captured data are not accepted without verification. The Scene Review screen must summarize the number of frames, coverage, detections, and capture duration so the officer can decide whether to proceed or capture the scene again. After this, the Incident Documentation form must properly combine AI-generated outputs with officer-entered information such as location description, vehicle

involvement, severity, road condition, tire evidence, casualties, officer details, and notes. The report generation and PDF export tests confirm that the finalized data can be converted into a complete accident report, while the completed record, read-only protection, data storage, and system log tests verify that the system preserves the integrity, traceability, and reliability of each case. Through these tests, ATADS can be assessed as a field-ready prototype that supports standardized, evidence-based, and more consistent traffic collision documentation.

Usability Test Plan

The Usability Test Plan evaluates whether the Automated Traffic Accident Documentation System (ATADS) can be used easily and correctly during the actual accident documentation process. Its main purpose is to check if the interface helps the officer move from one step to another without confusion, from opening the system, checking records, starting scene capture, reviewing AI-assisted outputs, completing the documentation form, and generating the final report. This is important because ATADS is intended to reduce common documentation problems such as missing measurements, unclear visual references, incomplete records, and inconsistent manual descriptions. A usable system must not only work technically; it must also help the user finish the task faster, with fewer mistakes, and with greater confidence.

User Entry and Welcome Screen

The welcome screen is tested to determine whether the user can immediately understand how to begin using ATADS. Since this is the first screen seen by the

officer, it must clearly show the system identity, basic instruction, and system condition without overwhelming the user with too much information. This is important because traffic accident documentation may happen in stressful roadside situations where the officer needs a fast and direct starting point. If the user can easily proceed from the welcome screen to the dashboard, it means the system provides a smooth entry point and avoids unnecessary delay at the beginning of the documentation workflow.

Incident Records Dashboard Usability

The incident records dashboard is tested because it serves as the main control center of ATADS. It must allow the user to quickly understand the number of total records, draft cases, records under review, completed reports, and individual incident details such as scene ID, date, location, collision type, vehicle involvement, and status. This is relevant because accident documentation does not end after capture; records must also be organized, monitored, reviewed, and retrieved. A clear dashboard helps the officer avoid confusion, reduces time spent searching for cases, and supports better management of accident records compared to scattered manual files or unorganized digital folders.

Search and Record Navigation

The search and record navigation function is tested to check whether the user can quickly locate a specific accident record using an incident ID, location, date, or

related keyword. This is important because ATADS may store multiple accident records, and officers must be able to retrieve the correct case without opening unrelated files. Accurate navigation protects the integrity of the documentation process because each record contains important details such as captured frames, AI analysis, officer notes, and generated reports. A usable search function helps users save time, avoid record mix-ups, and review previous cases more efficiently.

Unit Status and System Readiness

The unit status section is tested to ensure that the user can clearly understand whether the ATADS device is ready for field documentation. The system relies on key components such as the GPS, IMU, camera, and AI models, so their readiness must be visible before scene capture begins. This is important because incomplete sensor readiness may affect the quality of the accident record, especially if location data, orientation readings, or visual detection outputs are missing. A clear unit status display helps the officer make informed decisions before starting the capture process and prevents unreliable documentation from being collected unknowingly.

Live Capture Interface Usability

The live capture interface is tested because it is the main screen used for collecting accident scene evidence. It must clearly show the live camera feed, incident ID, unit number, GPS status, timer, detection count, and AI-assisted scene markers. This is important because the officer must be able to monitor the scene while the system identifies important accident elements such as vehicle position, rollover, debris, obstruction, point of contact, impact geometry, and elevation. A

usable live capture screen supports faster and more complete evidence collection by guiding the officer visually, instead of relying only on memory, rough estimates, or manual sketching.

AI Output Understanding

The AI output section is tested to determine whether the user can understand the system-generated findings and use them as support during documentation. ATADS presents outputs related to vehicle positioning, debris field, obstruction mapping, collision type, point of contact, impact geometry, and elevation data. This is important because AI findings are only useful if the officer can read and interpret them correctly. The outputs must be organized, properly labeled, and understandable in basic terms so that the user can use them as guidance, while still applying professional judgment when completing the final accident record.

Scene Review Usability

The scene review screen is tested because it acts as a quality checkpoint after live capture. It must show the captured frames, coverage percentage, detection count, and duration so the officer can decide whether the scene data are complete enough to proceed. This is important because poor or incomplete capture results can weaken the final report. The user must clearly understand when to select **Capture Again** and when to select **Proceed**. A usable review screen helps prevent incomplete visual evidence from moving forward and gives the officer control over the quality of the documentation before the incident form is completed.

Incident Documentation Form Usability

The incident documentation form is tested to check whether the user can complete the official accident record using both AI-assisted outputs and officer-entered details. The form must make it easy to enter or confirm important information such as scene ID, date and time, GPS coordinates, heading, location description, vehicle involvement, accident severity, incident status, traffic condition, road surface, roadway dimensions, tire evidence, casualties, officer name, badge number, and notes. This is relevant because the final report depends on complete and correct information. A usable form reduces missing details, organizes field observations, and supports more consistent traffic accident documentation.

Standardized Input Controls

Standardized input controls such as dropdowns, radio buttons, and fixed selection fields are tested because they help reduce unclear and inconsistent descriptions. Instead of requiring the officer to write every detail manually, ATADS allows the user to select common categories for severity, incident status, traffic condition, road surface, roadway dimensions, and vehicle involvement. This is important because standardized inputs make records easier to compare, review, and process. A usable set of input controls helps the officer complete the form faster while still maintaining accuracy and consistency in the accident documentation.

Required Field Awareness and Error Prevention

Required field awareness is tested to ensure that the system helps users avoid submitting incomplete accident records. ATADS must clearly show which fields are required and must prevent report generation if important information is missing.

This is important because one of the main weaknesses of manual documentation is the possibility of omitted measurements, missing descriptions, or incomplete case details. A usable error prevention system should not only block incomplete submission but also guide the user on what needs to be corrected. This makes the system supportive rather than frustrating and helps strengthen the completeness of the final report.

Report Generation Usability

The report generation function is tested to determine whether the user can easily complete the final step of the documentation process. After the officer verifies the AI outputs and fills in the required information, the **Generate Report** button must be easy to find and clearly understood as the action that creates the final accident documentation output. This is important because the report is the main deliverable of ATADS. A usable report generation process helps ensure that captured evidence, AI-assisted findings, and officer-verified details are properly combined into a complete and standardized accident report.

Completed Record Review and PDF Export

The completed record review and PDF export functions are tested to check whether the user can view finalized accident records and produce a formal copy when needed. A completed record must show the captured images, AI analysis, officer parameters, record status, and export option in a clear layout. This is important because completed records may be reviewed, filed, or used as reference

for further investigation. The read-only view also helps protect finalized information from accidental changes. A simple and visible **Export PDF** function helps ensure that the officer can prepare the report for submission or review without needing additional technical assistance.

Overall Workflow Usability

The overall workflow is tested to determine whether the user can complete the entire ATADS process from startup to final report without repeated assistance. This includes entering the system, viewing the dashboard, checking unit readiness, capturing the scene, reviewing results, completing the form, generating the report, viewing the completed record, and exporting the PDF. This is important because ATADS must work as one connected documentation process, not as separate screens. A successful workflow means the system is logical, easy to follow, and practical for field use, helping officers produce more complete and consistent accident records.

Usability Evaluation Criteria

The usability of ATADS will be evaluated based on clarity, navigation, task completion, error prevention, and user confidence. Clarity checks whether labels, buttons, AI outputs, forms, and status indicators are easy to understand. Navigation checks whether users can move from one screen to another without confusion. Task completion checks whether users can finish the documentation process successfully. Error prevention checks whether the system helps avoid missing or incomplete data. User confidence checks whether the officer feels comfortable using ATADS as a

documentation tool. These criteria are important because a system can only be effective if users can operate it correctly, efficiently, and with trust in its outputs.

Acceptance Test Plan

The Acceptance Plan defines how the Automated Traffic Accident Documentation System (ATADS) will be judged as acceptable for prototype approval. It focuses on whether the system can support the intended accident documentation workflow: opening the system, managing incident records, capturing accident scenes, reading GPS and IMU data, showing AI-assisted outputs, allowing officer verification, generating reports, and storing records properly. This plan is important because ATADS is designed to address manual documentation problems such as missing measurements, scale distortion, unclear orientation markers, ambiguous vehicle damage notation, and inconsistent landmark references. For the prototype to be accepted, it must show that it can provide a clearer, more organized, and more standardized way of documenting traffic collision scenes.

Acceptance Scope

The acceptance scope covers the major parts of ATADS that are directly connected to traffic accident documentation. These include the welcome screen, incident records dashboard, unit status panel, live capture screen, scene review screen, incident documentation form, completed record view, report generation, PDF export, database storage, and system logs. The acceptance process does not only check if the interface looks complete; it also checks whether the prototype can connect each stage into one continuous workflow. This is necessary because ATADS is intended to function as a field-support documentation tool where visual

evidence, sensor readings, AI outputs, officer inputs, and final reports must work together.

Acceptance Users and Evaluators

The acceptance process should involve the project evaluators, researchers, and intended system users such as traffic investigators or assigned officers who understand accident documentation needs. Their role is to check whether the prototype is understandable, practical, and aligned with the actual documentation process. The evaluators should focus on whether the system can guide a user from capture to report generation without unnecessary confusion. This is important because ATADS is not only a technical prototype; it is also a user-centered tool that must support officers working under time pressure, stress, and field conditions.

Workflow Acceptance

ATADS can be accepted if the full workflow can be completed from start to finish without major interruption. The user must be able to enter the system, view incident records, check unit readiness, start a capture session, review captured frames, proceed to the documentation form, complete officer inputs, generate a report, view the completed record, and export the PDF. This workflow is important because the system is designed to reduce documentation gaps at the source. If the prototype cannot guide the user through a complete and logical process, it cannot fully support its purpose as a standardized accident documentation system.

Interface Acceptance

The interface can be accepted if each screen is clear, readable, and easy to understand. The welcome screen must provide a simple entry point, the dashboard must organize accident records properly, the live capture screen must show the scene and system indicators clearly, the review screen must summarize captured data, and the documentation form must separate AI-assisted outputs from officer-entered details. This is important because the Technology Acceptance Model used in the design emphasizes that users are more likely to adopt a system if it is useful and easy to operate. ATADS must therefore present information in a way that reduces user confusion instead of adding technical burden.

Field Unit Readiness Acceptance

The field unit can be accepted if the system clearly shows the status of its main components before capture begins. The GPS must show whether location data are available, the IMU must show whether orientation data are ready, the camera must show whether visual capture is active, and the AI models must show whether detection and analysis features are loaded. This acceptance point is important because ATADS depends on these components to produce reliable documentation. If the officer cannot confirm device readiness, the captured accident data may become incomplete or less reliable.

Live Capture Acceptance

The live capture function can be accepted if it can display the accident scene clearly and show relevant AI-assisted markers during the capture process. These markers should help identify important accident elements such as vehicle position,

collision type, point of contact, debris field, obstruction, impact geometry, and elevation. This function is important because ATADS aims to reduce dependence on rough manual sketches by supporting accident scene documentation with visual capture, sensor data, and automated detection. A successful live capture process shows that the system can help gather scene evidence in a more guided and structured manner.

Scene Review Acceptance

The scene review function can be accepted if it allows the officer to check the captured results before moving to the documentation form. The screen must show the captured frames, coverage, detection count, and capture duration so the user can decide whether the evidence is complete enough. The system must also provide a clear option to capture again when the output is incomplete or unclear. This acceptance point is important because the review stage acts as a quality gate. It prevents weak or incomplete scene data from being used in the final report.

AI-Assisted Output Acceptance

The AI-assisted output can be accepted if the system presents its findings in an organized and understandable way. The outputs should include camera-based scene findings, GPS-based impact geometry, and IMU-based elevation or orientation information. These findings must support the officer by showing structured details such as vehicle positioning, debris field, obstruction mapping, collision type, point of contact, impact geometry, and z-axis elevation. This is important because raw image capture alone is not enough; ATADS must help convert captured scene data into useful accident documentation details.

Officer Input Acceptance

The officer input section can be accepted if users can complete all important accident details using clear fields, dropdowns, radio buttons, and text areas. The form must allow the officer to enter or confirm information such as location description, vehicle involvement, accident severity, incident status, traffic condition, road surface, roadway dimensions, tire evidence, casualties, officer name, badge number, and notes. This is important because ATADS still preserves the role of the officer in verifying and completing the accident record. The system assists the officer, but the final documentation must still include human-confirmed details.

Report Generation Acceptance

The report generation function can be accepted if the system can produce a complete documentation output after the officer verifies the required information. The generated report must combine the accident diagram, measured or verified parameters, officer-confirmed attributes, captured scene evidence, and related metadata. This stage is important because the final output represents the main deliverable of ATADS. The report generation phase must show that the prototype can transform captured data and officer inputs into a formal accident documentation package that can be reviewed, filed, and referenced.

Completed Record and PDF Export Acceptance

The completed record function can be accepted if finalized incident records can be reviewed clearly and exported as PDF. The completed view must show the captured scene, AI analysis, officer parameters, incident status, and export option. It

should also protect finalized details from accidental changes through read-only viewing. This acceptance point is important because completed records must remain consistent after verification. A clear PDF export function also supports the practical need to produce formal documentation for review and filing.

Database and Traceability Acceptance

The database structure can be accepted if it properly stores and connects the major records needed by ATADS. These include officers, incidents, capture sessions, scene frames, AI detections, AI analysis, reports, and system logs. This is important because the system must not only display information on the screen; it must preserve the documentation trail behind each accident record. Traceability allows each report to be connected to the responsible officer, captured evidence, AI outputs, generated reports, and recorded system activities.

System Log Acceptance

The system log function can be accepted if important system activities are recorded clearly. These activities include system startup, GPS lock, AI model loading, capture start, capture stop, record creation, report generation, and system errors. This is important because logs provide a technical history of the prototype's operation. They help evaluators review whether the system performed properly during testing and help identify possible issues when something does not work as expected.

Acceptance Criteria

ATADS will be considered acceptable if it can complete the full documentation process, present clear screens, monitor unit readiness, capture and review scene evidence, display understandable AI-assisted outputs, accept officer-verified information, generate reports, export completed records, and store all related data in a traceable structure. The prototype must also support the goal of reducing documentation omissions and improving standardization in traffic collision scene records. If these conditions are met, the system can be accepted as a research-ready prototype for improving accident documentation.

Final Acceptance Decision

The final acceptance decision should be based on whether ATADS successfully demonstrates its intended purpose as a guided, standardized, and evidence-supporting accident documentation system. If the prototype can help users complete the process from scene capture to final report generation with clear navigation, reliable data organization, and understandable outputs, it may be accepted for prototype evaluation. If major parts of the workflow fail, such as capture, review, documentation, report generation, or record storage, the system should be revised before final acceptance.

Conclusion

The development of the **Automated Traffic Accident Documentation System (ATADS)** presents a structured and technology-supported approach to improving traffic collision scene documentation. The prototype was designed to address common problems found in manual accident documentation, such as missing measurements, unclear sketches, scale distortion, incomplete references,

and inconsistent interpretation of scene details. Through the integration of camera capture, GPS data, IMU readings, AI-assisted detection, officer input, and report generation, ATADS provides a more organized method for recording accident scenes. This makes the system relevant because it supports a clearer and more standardized documentation process from scene capture up to final report preparation.

Response to Manual Documentation Problems

ATADS directly responds to the limitations of manual sketching and handwritten accident records. In traditional documentation, investigators may experience difficulty in recording exact vehicle positions, road conditions, debris fields, tire evidence, impact points, and scene orientation, especially when working under time pressure. These issues can lead to incomplete or unclear accident records. ATADS helps reduce these problems by guiding the user through a digital workflow where important scene details are captured, reviewed, and stored systematically. This improves the reliability of the documentation because the system encourages complete data collection before the final report is generated.

Importance of Sensor and Camera Integration

The use of camera, GPS, and IMU components strengthens the accuracy of the accident documentation process. The camera supports visual capture of the accident scene, the GPS records location-based information, and the IMU helps provide orientation and elevation-related readings. These components are important because accident scenes are physical spaces that require proper positioning,

direction, and measurement references. By combining visual evidence with sensor-based data, ATADS supports a more dependable accident record and reduces the risk of relying only on estimated measurements or memory-based observations.

Role of AI-Assisted Analysis

The AI-assisted features of ATADS provide additional support in identifying and organizing accident-related elements. The system helps detect and present information such as vehicle positioning, debris field, obstruction mapping, collision type, point of contact, impact geometry, and elevation data. These outputs help the officer understand the scene more clearly and complete the documentation form with better guidance. However, the system still preserves the role of the officer by allowing human verification and input. This is important because the prototype is not designed to replace investigators, but to assist them in producing more complete and consistent accident records.

Value of the Guided Workflow

The ATADS workflow is important because it arranges the documentation process into clear stages. The user begins at the welcome screen, proceeds to the incident records dashboard, starts a live capture, reviews the captured scene, completes the incident documentation form, and generates the final report. Each stage has a specific purpose and helps prevent the user from skipping important information. This guided process supports better documentation discipline because accident data are not simply entered randomly; they are collected, checked, completed, and finalized in an organized sequence.

Contribution of the Database Structure

The database structure supports the reliability and traceability of ATADS. It connects important records such as officers, incidents, capture sessions, scene frames, AI detections, AI analysis, reports, and system logs. This structure helps ensure that every accident record can be traced from the responsible officer to the captured images, AI findings, completed documentation, and generated report. This is important because traffic accident documentation must be accountable and reviewable. A well-organized database also helps prevent data loss, record confusion, and incomplete case tracking.

Significance of Report Generation

The report generation feature is one of the most important outputs of ATADS. It allows the system to convert captured evidence, AI-assisted findings, and officer-verified information into a structured accident report. This supports the goal of producing standardized documentation that can be reviewed, filed, and referenced when needed. Instead of relying only on separate images, handwritten notes, or manual sketches, ATADS brings the important accident details into one organized report package. This improves the usefulness of the system as a prototype for formal accident documentation.

Overall Contribution of ATADS

Overall, ATADS provides a strong foundation for improving traffic collision scene documentation through digital automation, structured data collection, and

guided user interaction. The system supports more complete records by combining visual capture, sensor readings, AI outputs, officer input, and database storage. Its contribution is not limited to making the process digital; it improves the quality of documentation by reducing missing information, improving consistency, and supporting evidence-based reporting. As a prototype, ATADS demonstrates how technology can assist investigators in creating clearer, clearer, and more organized accident records.

Final Statement

The development of ATADS shows that a guided and technology-assisted documentation system can help address the weaknesses of manual accident scene recording. By providing a structured workflow, sensor-supported capture, AI-assisted analysis, officer verification, and report generation, the prototype supports a more accurate and standardized approach to traffic collision documentation. With further testing and refinement, ATADS can serve as a practical foundation for improving how accident scenes are documented, reviewed, and preserved for future investigation and reporting.

Bibliography

Abohassan, M., & El-Basyouny, K. (2024). *Leveraging LiDAR-Based Simulations to Quantify the Complexity of the Static Environment for Autonomous Vehicles in Rural Settings*. MDPI. <https://www.mdpi.com/1424-8220/24/2/452>

Abohassan, M., & El-Basyouny, K. (2024, January 11). Leveraging LiDAR-Based Simulations to Quantify the Complexity of the Static Environment for Autonomous Vehicles in Rural Settings. *Multi-modal Sensor Fusion and 3D LiDARs for Vehicle Applications*, 24(2), 452.

<https://doi.org/10.3390/s24020452>

Ait-Moula, A., Riahi, E., & Serre, T. (2024, February 29). Effect of advanced rider assistance system on powered two wheelers crashes. *10(4)*.

doi.org/10.1016/j.heliyon.2024.e26031

Asmawi, M. Z., Mahamod, L. h., Mohamed, M. Z., & Paiman, T. (2014, July).

SUSTAINABLE GOVERNANCE IN RELATION TO THE FINANCIAL ASPECT IN MANAGING COASTAL AREAS: MALAYSIAN EXPERIENCE.

[10.21837/pmjournal.v12.i3.129](https://doi.org/10.21837/pmjournal.v12.i3.129)

Bagabaldo, A. R., & Hackl, J. (2025, October). DIGITAL TWINS FOR INTELLIGENT INTERSECTIONS: A LITERATURE REVIEW. <https://arxiv.org/pdf/2510.05374>

Brunken, H., & Gühmann, C. (2020, November 18). Road Surface Reconstruction by Stereo Vision. *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science*, 88. <https://doi.org/10.1007/s41064-020-00130-z>

Dai, F., Park, M.-W., Sandidge, M., & Brilakis, I. (2014, November). A vision-based method for on-road truck height measurement in proactive prevention of collision with overpasses and tunnels. *Automation in Construction*.

<https://doi.org/10.1016/j.autcon.2014.10.005>

Eskandarian, A., & Abbas, M. (2024, April). IEEE Transactions on Intelligent Transportation Systems (ITS) Special Issue on IEEE Workshop on Connected and Autonomous Vehicles. *IEEE Transactions on Intelligent Transportation Systems*. [10.1109/TITS.2024.3378808](https://doi.org/10.1109/TITS.2024.3378808)

- Farooq, D., & Juhasz, J. (2019, April). Simulation Analysis of Contributing Factors to Rider Visibility Issues for Car-Motorcycle Accidents. *Periodica Polytechnica Transportation Engineering*. 10.3311/PPtr.13521
- Guo, R., Liu, S., He, Y., & Xu, L. (2022, September 17). Study on Vehicle–Road Interaction for Autonomous Driving. <https://doi.org/10.3390/su141811693>
- Herbers, E., Doerzaph, Z., & Stowe, L. (2024, January). The Impact of Line-of-Sight and Connected Vehicle Technology on Mitigating and Preventing Crash and Near-Crash Events. 24(2). <https://doi.org/10.3390/s24020484>
- Jia, X., Zhang, C., & Jiang, Y. (2025, December 7). Spatial Retrieval Augmented Autonomous Driving. *Computer Vision and Pattern Recognition*. 10.48550/arxiv.2512.06865
- Jia, X., Zhang, C., Jiang, Y., & Wong, S. (2025, December). Spatial Retrieval Augmented Autonomous Driving. 10.48550/arXiv.2512.06865
- Kan, X., Zhu, S., Zhang, Y., & Qian, C. (2023, November). A Lightweight Human Fall Detection Network. doi.org/10.3390/s23229069
- Li, S., & Yoon, H.-S. (2023, March). Vehicle Localization in 3D World Coordinates Using Single Camera at Traffic Intersection. *Applications of Machine Learning in Automotive Engineering*. <https://doi.org/10.3390/s23073661>
- Loce, R. P., Bernal, E. A., Wu, W., & Bala, R. (2013, December). Computer vision in roadway transportation systems: a survey. 22(4). <https://doi.org/10.1117/1.JEI.22.4.041121>
- Ma, C., Zhou, J., & Yang, D. (2020, February). Causation Analysis of Hazardous Material Road Transportation Accidents Based on the Ordered Logit Regression Model. *Int. J. Environ. Res. Public Health* 2020, 17(4). <https://doi.org/10.3390/ijerph17041259>

- Mamčič, S., & Sivilevičius, H. (2013, April). The analysis of traffic accidents on Lithuanian regional gravel roads. 10.3846/16484142.2013.782894
- Markkula, G., Benderius, O., & Wolff, K. (2012, June). A Review of Near-Collision Driver Behavior Models. 54(6). doi/10.1177/0018720812448474
- Nitsche, P., Welsh, R. H., Genser, A., & Thomas, P. D. (2018, December). A novel, modular validation framework for collision avoidance of automated vehicles at road junctions. *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*. 10.1109/ITSC.2018.8569631
- OECD. (1999). Safety Strategies for Rural Road. *Road Transport and Intermodal Linkages Research Programme*. <https://doi.org/10.1787/9789264172913-en>.
- Perez, R. C. L. (2023, August 14). Data Mining technique: Application of Apriori algorithm for road accident analysis. *HCMCOUJS-Engineering and Technology*. 10.46223/HCMCOUJS.tech.en.13.2.2831.2023
- Rasol, M., Schmidt, F., Ientile, S., Adelaide, L., Nedjar, B., Kane, M., & Chevalier, C. (2021, September). Progress and Monitoring Opportunities of Skid Resistance in Road Transport: A Critical Review and Road Sensors. <https://doi.org/10.3390/rs13183729>
- Razi, A., Chen, X., Li, H., Wang, H., Russo, B., Chen, Y., & Yu, H. (2022, July). Deep Learning Serves Traffic Safety Analysis: A Forward-looking Review. *Computer Vision and Pattern Recognition*. <https://doi.org/10.48550/arXiv.2203.10939>
- SEO, Y., PARK, J., OH, G., KIM, H., HU, J., & SO, J. (2023, November). Text Classification Modeling Approach on Imbalanced-Unstructured Traffic Accident Descriptions Data. 10.1109/OJITS.2023.3335817

- Sun, L., Chen, L., Yin, Y., Tian, Y., & Zhang, X. (2021, December). Risk Assessment of Rollover and Skidding due to Pavement Roughness and Differential Settlement for Enhancing Transportation Safety. *Journal of Advanced Transportation*. <https://doi.org/10.1155/2021/7244283>
- Tabunar, S. M. S. (2020, October). KNOWLEDGE, ATTITUDE AND PRACTICES TOWARD ROAD SAFETY AMONG MEDICAL STUDENTS OF THE UNIVERSITY OF THE PHILIPPINES-COLLEGE OF MEDICINE. [10.37268/mjphm/vol.20/no.2/art.199](https://doi.org/10.37268/mjphm/vol.20/no.2/art.199)
- Tautkus, A., & Jurkauskas, A. (2003, April). THE ESTABLISHMENT OF CIRCUMSTANCES AND EVIDENCES OF AN ACCIDENT AND THEIR APPLICATION IN RESEARCH. *Transport*, 18(2). [10.3846/16483840.2003.10414069](https://doi.org/10.3846/16483840.2003.10414069)
- Tofighi, M., Asgary, A., Tofighi, G., Podloski, B., Cronemberger, F., Mukherjee, A., & Liu, X. (2021, November 25). Applying Machine Learning Models to First Responder Collisions Beside Roads: Insights from “Two Vehicles Hit a Parked Motor Vehicle” Data. *Appl. Sci.* 2021, 11(23). <https://doi.org/10.3390/app112311198>
- Wang, Y., Liang, Z., He, Y., Wu, J., Tian, P., & Ling, Z. (2025, July 28). AMFormer-based framework for accident responsibility attribution: Interpretable analysis with traffic accident features. <https://doi.org/10.1371/journal.pone.0329107>
- Wei, S., Shen, X., Shao, M., & Sun, L. (2021, November). Applying Data Mining Approaches for Analyzing Hazardous Materials Transportation Accidents on Different Types of Roads. <https://doi.org/10.3390/su132212773>

WON, M. (2020, April). Intelligent Traffic Monitoring Systems for Vehicle

Classification: A Survey. 10.1109/ACCESS.2020.2987634

Yang, X. L., Li, P., Lv, T., & Liao, X. H. (2013, September). Traffic Accident

Reconstruction Technology Research. *Advanced Materials Research*.

<https://doi.org/10.4028/www.scientific.net/AMR.756-759.946>

Zadobrischi, E. (2022, October). Intelligent Traffic Monitoring through

Heterogeneous and Autonomous Networks Dedicated to Traffic Automation.

<https://doi.org/10.3390/s22207861>

Curriculum Vitae

PROPOSERS' BASIC INFORMATION

Name: Law Justin A. Cosep

Position: BSCpE - 4 Student

Mobile/Landline: +639981552932

Email: lawjustin.cosep.22@usjr.edu.ph

Office/Department: Computer Engineering Dept.



EDUCATIONAL BACKGROUND

College: **University of San Jose - Recoletos**
2022 - Present

Senior High School: **St. Alphonsus Catholic School**
2020 - 2022

Junior High School: **St. Alphonsus Catholic School**
2016 - 2020

Elementary: **St. Alphonsus Catholic School**
2010 - 2016

RESEARCH ENGAGEMENTS/EXPERIENCES

Senior High School:

- THE EFFECTIVITY OF LAUC2 ANTENNA BY LAU HAN CHING AS AN ALTERNATIVE SIGNAL BOOSTER

PROPONENTS' BASIC INFORMATION

Name: Ralph James M. Lima
Position: BSCpE - 4 Student
Mobile/Landline: +639686160238
Email: ralphjames.lima.22@usjr.edu.ph
Office/Department: Computer Engineering Dept.



EDUCATIONAL BACKGROUND

College: **University of San Jose - Recoletos**
2022 - Present

Senior High School: **Passionist Sisters' School, Inc.**
2020 - 2022
With Honor

Junior High School: **Passionist Sisters' School, Inc.**
2016 - 2020
With Honor

Elementary: **Bawo Elementary School**
2010 - 2016
With Honor

RESEARCH ENGAGEMENTS/EXPERIENCES

Senior High School:

- AWAY FROM HOME: THE LIVED EXPERIENCE OF OVERSEAS FILIPINO WORKERS

PROPONENTS' BASIC INFORMATION

Name: Hannah Lyn T. Santiago
Position: BSCpE - 4 Student
Mobile/Landline: +639393388419
Email: hannahlyn.santiago.22@usjr.edu.ph
Office/Department: Computer Engineering Dept.



EDUCATIONAL BACKGROUND

College:	University of San Jose - Recoletos 2022 - Present
Senior High School:	University of Cebu - Main 2020 - 2022 With Honor
Junior High School:	Abellana National School 2016 - 2020 With Honor
Elementary:	Lahug Elementary School 2010 - 2016 With Honor

RESEARCH ENGAGEMENTS/EXPERIENCES

Senior High School:

- EFFECTS OF ACADEMIC PROCRASTINATION TO THE ACADEMIC PERFORMANCE OF STEM STUDENTS IN UNIVERSITY OF CEBU - MAIN